



Spatiotemporal patterns of ecological condition under water-salt stress in the yellow river delta: an improved remote sensing ecological index approach

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Abstract

Accurately assessing ecological condition is crucial for ecosystem conservation. The Remote Sensing Ecological Index (RSEI), which incorporates soil moisture (WET), was widely used to assess regional ecological condition. However, in coastal wetlands, the performance of the WET index is often undermined due to high soil salinity caused by seawater intrusion. Therefore, a Water-Salt Stress (WSS) index was developed by integrating WET and salinity (SI) to quantify the combined water-salt stress on vegetation. Replacing WET with WSS, an improved index, Water-Salt Stress Remote Sensing Ecological Index (WSRSEI), was established for ecological condition assessment. Results show higher WSS (>0.7) in tide-influenced coastal areas and lower WSS (<0.4) near inland channels. In high-WSS zones, WSRSEI was significantly lower than RSEI. Independent field validation demonstrate that WSRSEI significantly outperforms RSEI in capturing actual vegetation conditions. Long-term analysis (1985–2025) revealed a slight overall decline in the delta's WSRSEI (from 0.631 to 0.621). Ecological condition improved in the Qingshuigou Nature Reserve (QSGNR) (WSRSEI: 0.565 to 0.627) due to sediment input and post-2008 ecological water supplementation, but declined in the Diaokou River Nature Reserve (DKRNR) (0.621 to 0.554) due to coastal erosion and saltwater intrusion. Engineering structures such as levees and farm dikes effectively mitigated seawater intrusion. Since 1985, the expansion of aquaculture ponds, bare land, and built-up areas has been a major factor in the overall decline of ecological condition. This study provides a refined methodological framework for coastal ecological condition assessment.

Keywords Water-salt stress index · Ecological condition · Remote sensing · Spatiotemporal evolution

Introduction

Estuaries, situated at the interface between rivers and oceans, are critical components of global geochemical cycles that link continents and seas (Walling and Fang 2003; Meybeck et al. 2005). They serve as essential habitats and breeding

grounds for diverse species, including migratory and endangered species (Huang et al. 2024b; Ryu et al. 2021; Li et al. 2026). Furthermore, estuarine wetlands filter pollutants, mitigate eutrophication, and buffer floods by absorbing and gradually releasing water (Venturini et al. 2015; He et al. 2019; Day et al. 2007). Healthy estuaries also support coastal economies through fisheries and tourism while directly contributing to regional development (Wang et al. 2016; Barbier et al. 2011). Therefore, preserving these ecosystems is crucial for maintaining biodiversity, water quality, disaster resilience, and sustainable development.

The Yellow River Delta sustains diverse species and a variety of ecosystem types, supporting abundant biological resources and valuable genetic material that confer significant ecological value. As a critical node in the provision of global ecosystem services, it is China's only largely pristine

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large-river delta and hosts the most extensive, intact, and youngest wetland ecosystem within the country's warm temperate zone (Fan et al. 2011). It forms an essential part of the unique ecological framework of the Yellow River Basin. However, the evolution of estuarine delta wetland ecosystems is a complex process influenced by multiple factors, including but not limited to water availability, soil salinity, temperature and human activities (Balke et al. 2011). The Yellow River Delta's wetland ecosystem faces unique challenges: newly formed lands are characterized by shallow soils and low elevation, rendering them highly sensitive to tidal intrusion. In this setting, maintaining suitable hydrological conditions is essential for ecosystem health. Historically, the shifting courses of the Yellow River's estuary served as a vital driver for the formation and development of the delta's wetland ecosystems. Since 1855, the river's tail channels have shifted 10 times, creating 11 distinct flow paths across the delta (Wang and Liang 2000), with each path typically lasting only about a decade. However, the current Qingshuigou flow path, stabilized since May 1976, has led to prolonged single-channel flow, limiting ecological water supplementation to the abandoned Diaokou River flow path. Additionally, although reservoir regulation has prevented complete river flow interruptions, increasing water scarcity and rising downstream diversions have eliminated the natural spring flood pulses known as "Taoxun". Instead, most water and sediment now rapidly discharge into the sea between late June and early July (Hu and Zhang 2009). Consequently, spring (March–May) water flows are critically low and insufficient to meet the replenishment demands of the wetland ecosystem (Ni et al. 2002; Liu et al. 2020). Such conditions not only impair normal wetland vegetation growth but may also trigger reverse ecosystem succession—a regression from complex advanced communities to simpler primary assemblages (Zhang et al. 2022; Huang et al. 2024a).

Wetland ecological condition assessment is crucial for maintaining ecosystem functions and services, serving as a key component of ecological conservation and restoration efforts. Early research primarily relied on field sampling and laboratory analyses, employing traditional approaches such as water quality testing, soil sample collection, and vegetation surveys (Fu et al. 2021). These methods yield high-precision data with clear mechanistic understanding, forming the foundation of ecological knowledge. However, they are time-consuming, labor-intensive, and costly, with extremely limited spatial coverage. Consequently, they struggle to capture the highly heterogeneous and dynamically changing macro-meso scale patterns of estuarine wetlands, and cannot meet the demands of long-term, continuous regional-scale monitoring. In recent years, process simulation assessments based on ecological models, represented by the

InVEST model, have been developed (Gomes et al. 2021). These approaches can incorporate disturbances from human activities such as roads, settlements, and farmland, as well as invasive species, offering the advantage of clear causal logic. They can be used to predict the impacts of future land use changes on habitats, demonstrating strong decision-support capabilities. However, the InVEST model relies heavily on expert experience or literature for parameters such as threat weights, sensitivity, and attenuation distances during habitat quality assessment, lacking universal standards. This affects the comparability of ecological model assessment results across different years, and the model also demands high precision in land use type data (Wei et al. 2022).

With advancements in environmental science, remote sensing technology, and geographic information systems, assessment methods for wetland habitat quality have undergone significant evolution (Jesudhas and Roy 2024). Comprehensive ecological assessment tools based on multi-source remote sensing data—represented by the Remote Sensing Ecological Index (RSEI)—have gained widespread application in ecological monitoring and evaluation across diverse geographical environments due to their advantages in data accessibility, objective weight determination, and result visualization. Initially proposed by Xu (2013), the RSEI integrates four key ecological factors—wetness, greenness, dryness, and heat—through Principal Component Analysis (PCA) to quantify regional ecological quality changes (et al., 2025). A major advantage of this model is its use of high-resolution remote sensing imagery, which facilitates ecological assessments with high spatiotemporal resolution and helps overcome limitations such as poor representativeness inherent in traditional ground-based surveys (Gan et al. 2024). As research progressed, limitations of the standard RSEI in capturing complex ecosystem traits have become apparent, prompting numerous refinements. Consequently, various improvements have been proposed to enhance its capabilities. For instance: Yang et al. (2023) introduced Net Primary Productivity (NPP) into habitat assessment in mining areas. NPP provides an estimate of actual carbon fixation based on photosynthetic processes, and its dynamic changes are often more sensitive than NDVI (Normalized Difference Vegetation Index) in environments with sparse vegetation or significant stress, thereby offering valuable supplementary information for habitat assessment at the functional level. The three-dimensional greenness index proposed by Liu et al. (2023) aims to enhance the assessment accuracy of RSEI in ecosystems with vertical structure characteristics. By integrating spatial structure information, this index can more finely characterize the vertical distribution of vegetation and canopy complexity, providing spatial details that traditional NDVI struggles to capture. Wang et al. (2025), in monitoring ecological restoration in open-pit mining areas,

comprehensively utilized the desertification index and red-edge NDVI. This improvement addresses the typical "bare soil-vegetation" mixed pixel problem in mining areas and the need for early identification of vegetation stress. Leveraging the characteristic that vegetation is more sensitive to physiological stress in the red-edge band, red-edge NDVI can detect changes in vegetation physiological status earlier than traditional NDVI, enhancing the comprehensive monitoring capability of the ecological restoration process. These modified indexes adhere to a shared philosophy: preserving the original framework while introducing targeted optimizations to enhance regional applicability, thereby generally validating the ability of RSEI to capture ecological change trends on a large scale. In estuarine delta regions, periodic tidal inundation imposes significant hydrological and salinity stress on wetland vegetation (Feng et al. 2018), often triggering degradation and subsequent habitat loss. However, the standard RSEI model exhibits a critical flaw in such environments: its soil moisture component cannot differentiate between freshwater and saltwater. This limitation introduces substantial bias into the assessment, thereby compromising the model's accuracy and reliability in these ecologically vital areas.

To enhance the accuracy and applicability of the RSEI model in estuarine delta regions, this study improves upon the original RSEI framework by introducing a salinity index (SI) and combining it with the wetness index (WET) index to construct a Water and Salt Stress Index (WSS). This innovation leads to the development of a Water-Salt stress RSEI model (WSRSEI) that effectively captures ecological condition under water-salt stress conditions, thereby significantly improving the model's adaptability for estuarine delta ecosystems. The study then employs this enhanced WSRSEI model to assess ecological condition changes in the modern Yellow River Delta from 1985 to 2025. Finally, it systematically examines the evolution mechanisms of ecological condition in the delta by analyzing multiple driving factors, including: spit erosion and deposition dynamics, ecological water supplementation, artificial geomorphological changes, and land use transformations. This research provides valuable insights for understanding wetland habitat evolution mechanisms in estuarine deltas under multiple stressors, offering both methodological and practical contributions to coastal ecosystem management.

Study area

The Yellow River, widely known as one of the world's most sediment-rich rivers, originates from the Qinghai-Tibet Plateau and flows into the Bohai Sea. With a total length of 5,464 km and a drainage basin area of 752,000 km² (Fig. 1b),

it represents one of the most critical and dynamic fluvial systems in China. The Yellow River Estuary is situated at the confluence of the Bohai Sea and Laizhou Bay, serving as the primary discharge area for Yellow River water and sediment. It plays a vital role in the river's overall management, development, and governance. The modern Yellow River Delta (118°20'E–119°20'E, 37°16'N–38°16'N) forms a fan-shaped area with Yuwa as its apex, bounded by the Tiao River to the north and the Songchunrong Channel to the south (as delineated by the red fan-shaped boundary in Fig. 1a). The Qingshuigou course, the current Yellow River channel, has been in operation since May 1976 (Du et al. 2024) and remains active today, with over 40 years of channel stability. From 1976 to 1996, it flowed southeast into Laizhou Bay. In May 1996, to accommodate oilfield development, the Qing 8 diversion project redirected the channel northeast (Fig. 1b). The southeastern channel (which was active between 1976 and 1996) has since received no further water or sediment supply. The Diaokou River course, a former Yellow River channel, was active from January 1964 to May 1976 and now serves as a backup channel. Rich in oil and marine resources, this area suffered ecological degradation after the 1976 diversion cut off sediment and freshwater inputs, leading to coastal erosion, saltwater intrusion, and wetland loss. To protect the delta's wetlands and endangered species, the Shandong Yellow River Delta National Nature Reserve was established in 1992 (Zhang et al. 2017). The reserve covers both the active Qingshuigou and abandoned Diaokou River estuaries (green boundary in Fig. 1b). It hosts extensive natural and constructed wetlands, particularly freshwater marshes and intertidal zones that serve as critical bird habitats (Li et al. 2024).

Data and methods

Data sources

Remote sensing imagery

All remote sensing images used in this study were derived from the U.S. Geological Survey (USGS) Landsat series satellites and were accessed and preprocessed via the Google Earth Engine (GEE) cloud platform. Specifically, we utilized Landsat 5 Thematic Mapper (TM) imagery (1985–2011), Landsat 7 Enhanced Thematic Mapper Plus (ETM+) imagery (2012), and Landsat 8 Operational Land Imager (OLI) imagery (2013–2025). All images were obtained as Collection 2 Level-2 science products, which include atmospheric correction and surface reflectance retrieval as standard outputs. To ensure temporal comparability, we selected images acquired during the peak vegetation growing season

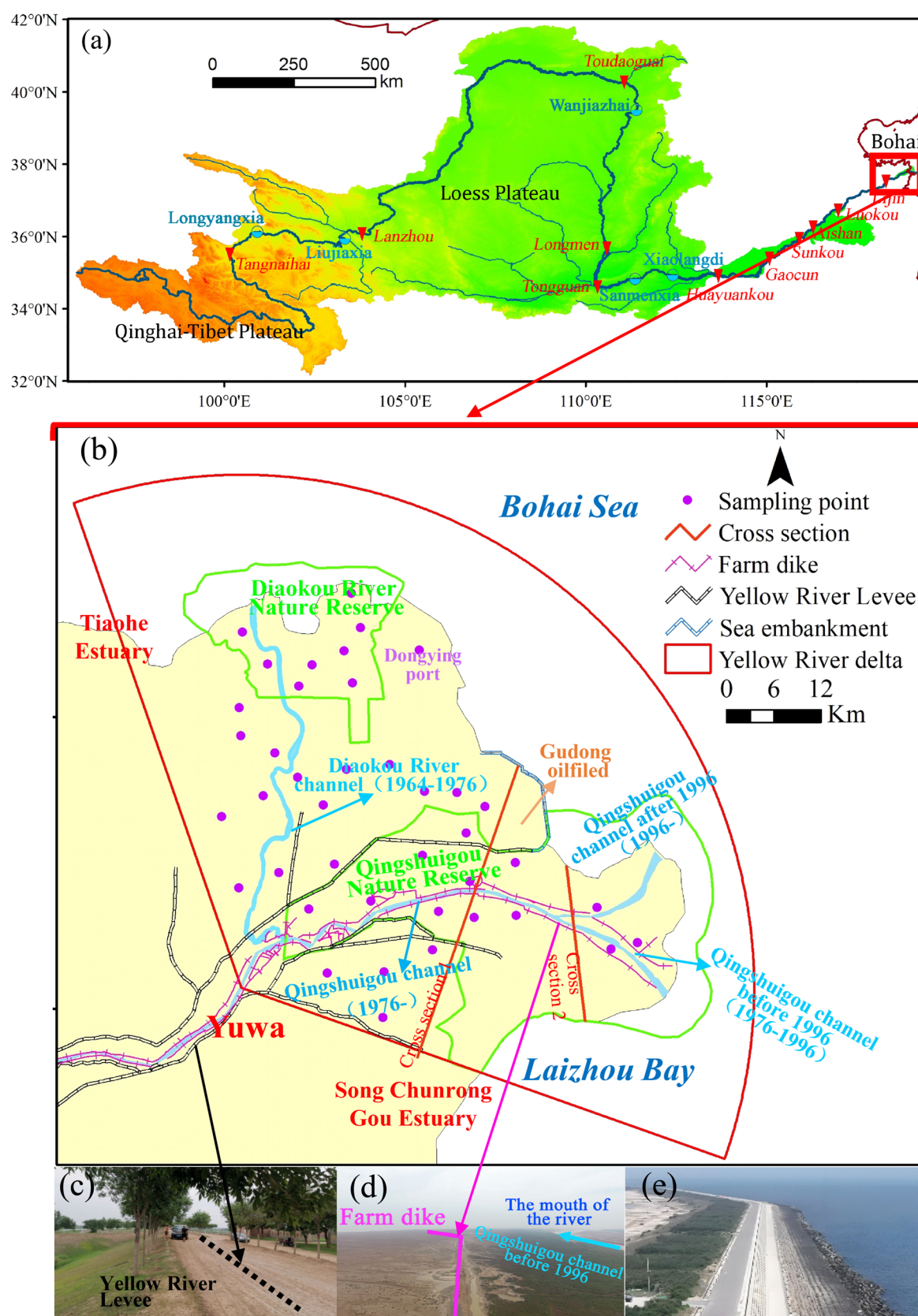


Fig. 1 **a** Spatial extent of the modern Yellow River Delta, showing channel distributions and locations of nature reserves; **b** Position of the Yellow River Estuary within the Yellow River Basin; **c** Main levees

of the Yellow River; **d** Distribution of vegetation and production dikes along the left bank of the Qingshuigou channel prior to 1996 (taken in June 2020); **e** Gudong Sea embankment

(June–September) for each year, prioritizing scenes with cloud cover less than 10%.

Cross section data

To analyze the influence of topography on the spatial distribution of water-salt stress and ecological conditions, we conducted topographic surveys along two representative cross-sections in the lower Qingshuigou channel during October 2019 (cross-section locations and parameters are provided in Table 1). Surveys were performed using an SDH-13D single-frequency echo sounder, with a measurement error of less than 5 cm. Survey points were spaced at approximately 100 m intervals along each transect, yielding a total of 660 valid elevation points. All elevation values were referenced to the 1985 National Elevation Datum of China.

The cross-section data were used to: (1) compare topographic differences between areas inside and outside the levees; (2) analyze the spatial correspondence between elevation and WSS/WSRSEI values in conjunction with concurrent remote sensing imagery; and (3) elucidate the mechanisms by which engineered geomorphic structures influence ecological condition.

land use data

Land use data from 1985 to 2020 were employed. The dataset used is GLC_FCS30D (Global Land Cover Fine Classification System 30m Dynamic), which provides global land cover maps at a 30 m spatial resolution. It employs a fine classification system containing 35 land-cover categories, and the data were aggregated into nine types for this study: Cropland, Grassland, Shrubland, Sparse vegetation, Marsh, Forest, Waterbody, Bare areas, and Built-up. The product covers the period from 1985 to 2020. The dataset was developed by combining continuous change detection from dense Landsat time-series imagery, locally adaptive updating models, and spatiotemporal optimization algorithms. Validation using the LCCS Level-1 system (17 land use classes) achieved an accuracy of 73.24% ($\pm 0.30\%$), while the overall accuracy for 10 major land use types reached 80.88% ($\pm 0.27\%$). The dataset is available at: <https://zenodo.org/records/8239305>. The detailed methodology is described in Zhang et al. (2024a, b). In this study, we applied the following preprocessing steps: (1) reprojection to a unified

coordinate system (WGS 1984 UTM zone 50N); and (2) clipping to the Yellow River Delta boundary.

Field validation data

A field campaign was conducted in the Yellow River Delta in June 2020, synchronized with Landsat overpass dates to ensure temporal alignment between remote sensing and ground measurements.

(1) Sampling site layout

Considering the water-salt gradient, vegetation type distribution, and accessibility of the Yellow River Delta, a total of 40 soil–vegetation co-located sampling sites were established within the study area (Fig. 1b).

(2) Soil sample collection and laboratory analysis

Soil sample collection: At each sampling site, surface litter was removed, and topsoil samples (0–20 cm) were collected using a stainless steel soil auger. Approximately 500 g of each sample was placed in sealed polyethylene bags, labeled, and refrigerated during transport to the laboratory. In addition, undisturbed soil cores were collected using a 100 cm³ cutting ring for soil moisture determination.

Soil salinity measurement: Soil samples were air-dried, ground, and passed through a 2 mm sieve. Soil extracts were prepared at a water-to-soil ratio of 5:1. After thorough shaking for 30 min, the suspensions were filtered, and the electrical conductivity (EC, dS/m) of the extracts was measured using a conductivity meter (DDS-307 or equivalent).

Soil moisture measurement: Soil gravimetric moisture content was determined using the oven-drying method. Approximately 20 g of fresh soil was placed in a pre-weighed aluminum tin and weighed to the nearest 0.01 g. The samples were then dried in an oven at $105 \pm 2^\circ\text{C}$ to constant weight (at least 8 h for clayey soils and 6 h for sandy soils). After drying, the samples were cooled in a desiccator to room temperature and reweighed. Soil moisture content (MSC) was calculated as:

$$MSC = \frac{M_{wet} - M_{dry}}{M_{dry}} \times 100\% \quad (2-1)$$

Table 1 Characteristics of the cross sections location

Cross section	Distance from Yuwa (km)	Section width(m)	Left endpoint coordinate		Right endpoint coordinate	
			X	Y	X	Y
Section 1	37.12	39317	680732	4202329	667847	4165183
Section 2	49.61	22409	686510	4192056	689203	4169809

where M_{wet} and M_{dry} are the masses (g) of wet and oven-dried soil, respectively. Each sample was analyzed in duplicate, and the arithmetic mean was taken as the final result, with the difference between the two measurements not exceeding 1%. Results are expressed as percentages (%).

(3) Vegetation quadrat survey

A 1 m × 1 m vegetation quadrat was established adjacent to each sampling site. Within each quadrat, above-ground vegetation was clipped at ground level, with litter removed. The samples were placed in labeled envelopes and transported to the laboratory, where they were oven-dried at 65–80°C for 12 h and weighed to determine the dry above-ground biomass (g/m²).

Methods

The detailed workflow is illustrated in Fig. 2. First, representative annual Landsat remote sensing images were selected and underwent atmospheric correction processing and cloud removal. Subsequently, various indexes

including NDVI, NDBSI (Normalized Difference Built-up and Soil Index), LST (Land Surface Temperature), WET, SI, and FVC (Fractional Vegetation Cover) were calculated. Based on WET, SI, and FVC, the water-salt stress index (WSS) was constructed. These indexes were then subjected to PCA to develop both the RSEI model and the WSRSEI model. Following this, the Modified Normalized Difference Water Index (MNDWI) was computed and the OTSU thresholding method was applied to distinguish water bodies from terrestrial areas, generating a water mask that was applied to remove water pixels from the RSEI, WSRSEI and all individual indices (NDVI, WET, NDBSI, LST, SI, FVC). Multi-year normalization was then applied to RSEI, WSRSEI and all individual indices, using only the remaining land pixels, to ensure comparability across years. On this basis, the applicability of WSRSEI and its advantages over RSEI in assessing ecological condition in estuarine regions were discussed. The WSRSEI model was then employed to analyze the spatiotemporal evolution characteristics of ecological condition in the modern Yellow River Delta from 1985 to 2025. Finally, the evolutionary mechanisms of estuarine wetland habitats were explored from perspectives including spit erosion/deposition, ecological water

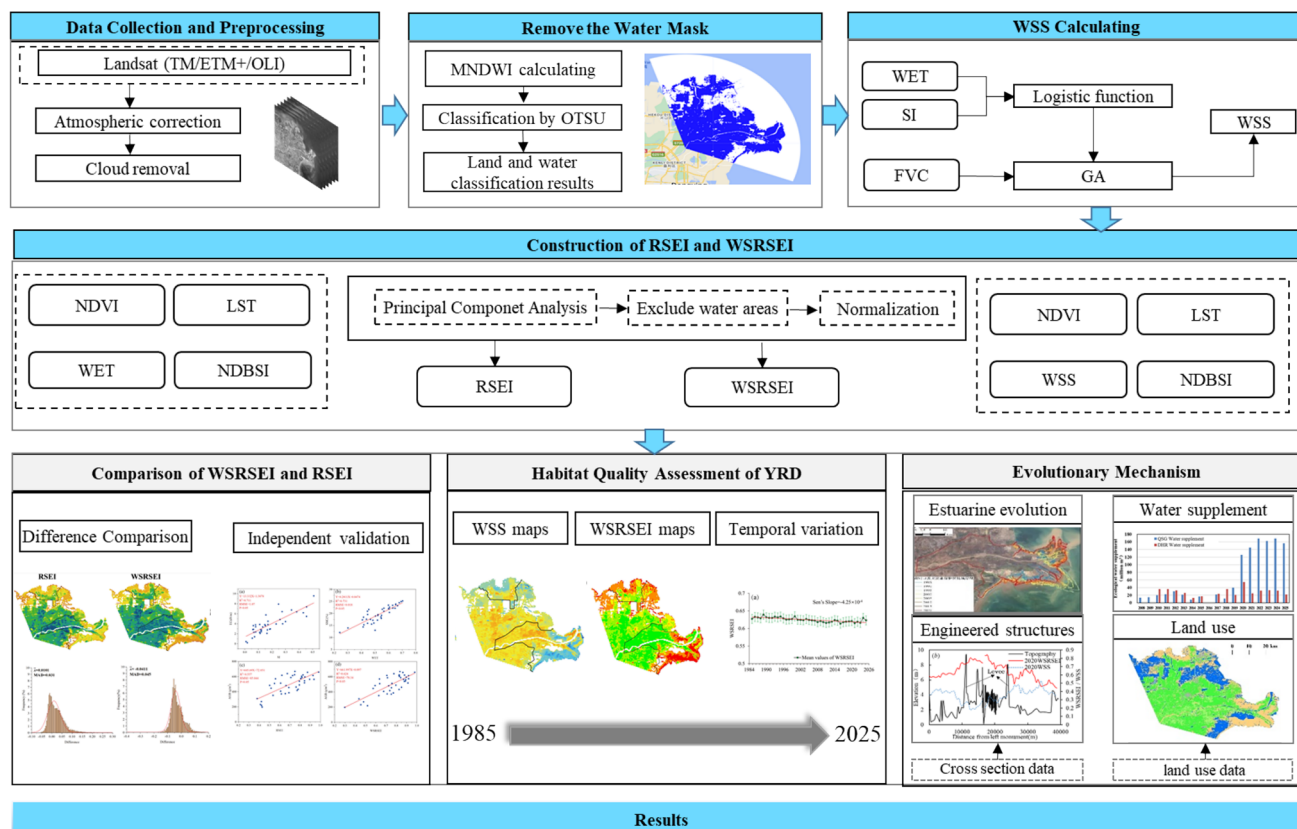


Fig. 2 Methodological workflow for the construction of RSEI and WSRSEI. The workflow includes index calculation, PCA transformation, water masking, normalization, and subsequent spatiotemporal analysis

supplementation, engineered geomorphological changes, and land use transformations. It should be noted that the WET index used in both RSEI and WSRSEI were represented by the wetness component of the Tasseled Cap transformation (which reveals soil and vegetation moisture), rather than conventional water index. Therefore, for each annual remote sensing image, marine and inland water areas were removed according to the specific conditions of the study area that year (Zhang et al. 2024a, b). The remote sensing images utilized in this study were all acquired during June–September each year, as this period

best reflects the ecological conditions during the regional wetland vegetation growing season.

Water-salt stress index

To quantify the nonlinear effects of soil moisture (WET, range: −1 to 1) and salinity (SI, range: 0 to 1) on wetland vegetation, a logistic function (Awad et al. 2023) was adopted:

$$WSS = \frac{1}{1 + \exp[a + b \cdot SI + c \cdot (WET + 1)]} \quad (3-1)$$

where a , b , and c are parameters optimized via a genetic algorithm (GA) in MATLAB (Shen et al. 2020) to maximize the coefficient of determination (R^2) between WSS and FVC. SI was calculated using the method summarized in Table 2, which has been specifically validated for the Yellow River Delta (Cheng et al. 2022).

The GA used for WSS parameter optimization was implemented in MATLAB with the following settings: population size=200, maximum generations=500, cross-over fraction=0.8, mutation rate=0.01, and elite count=2. The optimization was set to terminate when the change in the best fitness value over 50 consecutive generations fell below 1×10^{-6} , or when the maximum number of generations was reached.

The logistic function can describe the response characteristics of ecological variables that change rapidly at first and then tend to stabilize as ecological factors (such as soil salinity and moisture) increase. This aligns more closely with the actual mechanisms of resource limitation and threshold effects in ecosystems. Furthermore, studies have shown that when simulating the nonlinear effects of environmental factors on ecological processes, the logistic model is often more realistic and stable compared to traditional equations in many cases (Ma et al. 2021; Yu and Ye 2020).

The logistic function, as an S-shaped curve with both upper and lower bounds, has inherent advantages in simulating the nonlinear effects of environmental factors on biogeochemical processes (Yu & Ye 2020). The response of vegetation to water-salt stress is not linear: under low stress levels, vegetation maintains normal growth; when stress exceeds a certain threshold, vegetation declines rapidly; and under extreme stress, the decline rate gradually slows and stabilizes. The S-shaped curve of the logistic function precisely captures this three-phase pattern of "slow change → rapid response → saturation." In studies of salt marsh wetlands in the Yellow River Delta, the logistic function has been used to simulate population growth and interspecific competition of salt marsh

Table 2 Calculation method of *NDVI*, *WET*, *NDBSI*, *LST*, *SI* and *FVC*

Index	Calculation Formula	Explanation
<i>NDVI</i>	$(B_{nir} - B_{red}) / (B_{nir} + B_{red})$	B_{blue} , B_{green} , B_{red} , B_{nir} , B_{swir1} , B_{swir2} represent the spectral bands of Landsat 5 TM, Landsat 7 ETM+ and Landsat 8 OLI/TIRS, respectively; C_1 , C_2 , C_3 , C_4 , C_5 , C_6 denote the corresponding band coefficients for these sensors (Kodl et al. 2024) (Zhang et al. 2023)
<i>WET</i>	$C_1 B_{blue} + C_2 B_{green} + C_3 B_{red} + C_4 B_{nir} + C_5 B_{swir1} + C_6 B_{swir2}$	
<i>NDBSI</i>	$(IBI + SII) / 2$	<i>IBI</i> and <i>SII</i> respectively represent the building index and the bare land index. (Yue et al. 2019)
<i>LST</i>	$K_2 / \ln[K_1 B(T_s) + 1]$	$B(T_s)$ represents the blackbody radiance, while K_1 and K_2 denote the sensor-specific calibration coefficients for the thermal bands Landsat 5 TM, Landsat 7 ETM+, and Landsat 8 OLI/TIRS (Pande et al. 2024) (Fouad 2003)
<i>SI</i>	$(B_{swir1} B_{swir2} - B_{swir2} B_{swir1}) / B_{swir1}$	
<i>FVC</i>	$(NDVI - NDVI_{soil}) / (NDVI_{veg} - NDVI_{soil})$	$NDVI_{soil}$ represents the <i>NDVI</i> value of bare soil, $NDVI_{veg}$ denotes the <i>NDVI</i> value under complete vegetation cover (Song et al. 2022)

vegetation (Yi et al. 2021). A significant logistic relationship has been observed between the normalized difference vegetation index (NDVI) and groundwater chloride concentration, indicating that the logistic function can effectively characterize the stress effects of water-salt factors on vegetation (An et al. 2011). These prior applications confirm that the logistic function is an appropriate choice for characterizing vegetation response to water-salt stress in this region.

Calculation methods for RSEI and WSRSEI

The RSEI ecological condition assessment model integrates four indexes—NDVI, WET, NDBSI, and LST—through PCA (Huang et al. 2024a). Based on research requirements, principal components with higher eigenvalue contributions are retained to synthesize a new composite image, expressed as: $RSEI = f(NDVI, WET, NDBSI, LST)$. To ensure that higher pixel values in the composite image represent better ecological conditions, the initial RSEI is calculated using the following transformation:

$$RSEI = 1 - PC(f(NDVI, WET, NDBSI, LST)) \quad (3-2)$$

Similarly, the WSRSEI model combines four indexes—NDVI, WSS, NDBSI, and LST. It can be specifically expressed by the following formula:

$$WSRSEI = 1 - PC(f(NDVI, WSS, NDBSI, LST)) \quad (3-3)$$

Table 2 provides the detailed calculation formulas for all indexes.

Method for extracting land vector data

(1) Calculation of MNDWI

The MNDWI represents an enhancement of the NDWI (Normalized Difference Water Index) (Jiang et al. 2024), designed to improve the differentiation between water bodies and other land use types. The MNDWI is computed using the green band (Green) and mid-infrared band (MIR), with its formula expressed as:

$$MNDWI = \frac{(Green - MIR)}{(Green + MIR)} \quad (3-4)$$

The MNDWI demonstrates superior performance compared to the traditional NDWI in distinguishing water bodies from other land use types (e.g., bare soil and built-up areas).

(2) Adaptive Land–Water Binary Segmentation Algorithm: OTSU Method

OTSU method (Ren et al. 2024; Liu et al. 2024), also known as the maximum inter-class variance method, is an adaptive binarization algorithm that automatically determines the optimal threshold from an image's gray-scale histogram to segment the image into foreground (e.g., target objects) and background. The core principle involves maximizing the variance between two pixel classes (foreground and background) to identify the optimal segmentation point.

The between-class variance (σ^2), which measures the separation between foreground and background, is calculated as the squared difference of their mean gray values weighted by their respective pixel proportions:

$$\sigma_B^2(t) = w_0(t) \cdot w_1(t) \cdot [\mu_0(t) - \mu_1(t)]^2 \quad (3-5)$$

where: w_0 and w_1 represent the weights (i.e., proportion of total pixels) for the foreground and background classes of the MNDWI image, respectively; and μ_0 and μ_1 denote the mean gray values of the foreground and background classes, respectively.

The algorithm evaluates all possible threshold values and selects the one that maximizes $\sigma_B^2(t)$ as the final segmentation threshold. This optimal threshold ensures maximum discriminability between foreground and background classes, thereby achieving effective image segmentation.

Comparative analysis method for RSEI and WSRSEI

This study employs the Mean Absolute Difference (MAD) metric (Sun et al. 2022) to quantitatively evaluate discrepancies between RSEI and WSRSEI models across different regions. The MAD is calculated as:

$$MAD = \frac{1}{n} \sum_{i=1}^n |Y_i - X_i| \quad (3-6)$$

where: n represents total number of pixels in the study area; Y_i represents WSRSEI value of the i -th pixel; X_i represents RSEI value of the corresponding i -th pixel.

Uncertainty quantification method

To quantify the composite uncertainty of the annual mean WSRSEI, we evaluated three sources of uncertainty: multi-sensor consistency deviation (U_{sensor}), PCA loading inter-annual variability (U_{PCA}), and GA parameter optimization

uncertainty (U_{GA}). U_{sensor} was estimated from the differences in WSRSEI between overlapping observation periods of Landsat 5 and Landsat 7 (1999–2011) and Landsat 5 and Landsat 7 (2013–2020). U_{PCA} was derived from the interannual coefficient of variation of the first principal component loadings from 1985 to 2025. U_{GA} was estimated from the results of 20 independent GA runs. The composite standard error was synthesized using the root-sum-square method:

$$SE = \sqrt{U_{sensor}^2 + U_{PCA}^2 + U_{GA}^2} \quad (3-7)$$

Trend analysis and spatial statistics methods

To comprehensively characterize the spatiotemporal evolution of WSRSEI, we conducted analyses from three perspectives: trend significance, trend persistence, and spatial autocorrelation.

For trend significance, we combined the Mann–Kendall (M–K) trend test with Sen's slope estimation. The M–K test is a non-parametric statistical method that does not require normality assumptions and is insensitive to outliers, making it suitable for long-term trend significance testing. At a significance level of $\alpha=0.05$, a trend was considered statistically significant when $|Z|>1.96$. Sen's slope was calculated as the median of the slopes of all possible year-to-year combinations in the time series, providing a robust estimate of the trend magnitude and direction.

For trend persistence, we calculated the Hurst exponent to assess the future continuity of WSRSEI trends. The Hurst exponent H ranges from 0 to 1: $H>0.5$ indicates that future trends will continue in the same direction as the past (persistence), while $H<0.5$ suggests a reversal (anti-persistence). The Hurst exponent was estimated using the rescaled range (R/S) analysis method.

For spatial autocorrelation, we employed global Moran's I and local Moran's I (LISA) to evaluate the spatial clustering patterns of WSRSEI. Global Moran's I measures the overall spatial autocorrelation across the study area, while LISA identifies local clustering patterns.

Quantitative attribution analysis methods

To quantitatively assess the contributions of various driving factors to WSRSEI changes, we employed two statistical methods: Partial Least Squares Regression (PLSR) and independent samples t-test.

(1) Partial Least Squares Regression (PLSR)

PLSR is a multivariate regression method that handles multicollinearity among independent variables by

extracting orthogonal latent components that maximize the covariance between the independent and dependent variables. This method is particularly suitable for ecological attribution analysis where environmental drivers often exhibit strong intercorrelations (e.g., sediment load and land area change).

In this study, PLSR was applied to quantify the contributions of driving factors to WSRSEI trends in the two nature reserves. The dependent variable (Y) was the annual mean WSRSEI (1985–2025), and the independent variables (X) included land area and ecological water supplementation volumes. All variables were standardized (Z -score) prior to analysis to ensure comparability of regression coefficients.

The optimal number of latent components was determined by leave-one-out cross-validation, selecting the number that maximized the cross-validated R^2 (Q^2). Variable importance in the projection (VIP) scores were calculated to identify the relative importance of each independent variable, with $VIP>1.0$ considered as a key driver. The contribution rate (%) of each factor was calculated based on the absolute values of the standardized regression coefficients:

$$Contribution_i = \frac{|b_i|}{\sum_{j=1}^n |b_j|} \times 100\% \quad (3-8)$$

where (b_i) is the standardized regression coefficient of the i -th independent variable. Model performance was evaluated using the goodness-of-fit R^2 and the cross-validated Q^2 . A $Q^2>0.5$ was considered indicative of good predictive ability. All PLSR analyses were performed using the Python's 'scikit-learn' library.

(2) Independent samples t-test

To test whether WSRSEI differed significantly between areas inside and outside the levees, we employed an independent samples t-test. This method compares the means of two independent groups (inside vs. outside the levees) to determine whether the observed difference is statistically significant.

Prior to the t-test, the homogeneity of variances was assessed using Levene's test. If variances were equal ($p>0.05$), the standard t-test with pooled variance was used; otherwise, Welch's t-test (which does not assume equal variances) was applied. The normality assumption was relaxed given the sample size ($n>30$ in each group) and the robustness of the t-test to moderate deviations from normality.

The effect size was quantified using Cohen's d :

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}} \quad (3-9)$$

where $\bar{x}_1$ and $\bar{x}_2$ are the group means, and s_{pooled} is the pooled standard deviation. Effect sizes were interpreted as small ($d \approx 0.2$), medium ($d \approx 0.5$), or large ($d \geq 0.8$). Statistical significance was set at $\alpha = 0.05$. All t -tests were performed using Python's 'scipy.stats' module.

Results

Evolution characteristics of the water-salt stress index

The GA was employed to determine the parameters in Eq. (3-1), yielding optimal values of $a = 5.5$, $b = -13.2$, and $c = -4.68$, with the coefficient of determination (R^2) between WSS and FVC reaching 0.686 (Fig. 4j). This parameter configuration has clear ecological implications: in the Yellow River Delta, the stress contribution of salinity (SI) to vegetation is approximately 2.8 times that of moisture, confirming that soil salinization caused by seawater intrusion is the primary stressor for wetland vegetation in this region, while moisture stress is secondary. This finding is consistent with field observations of the "salinity-dominated" stress regime in the Yellow River Delta (An et al. 2011).

Parameter sensitivity analysis shows that when the optimal parameters were perturbed within $\pm 20\%$, the change in R^2 (ΔR^2) between WSS and FVC remained below 0.001 (Supplementary Table S2), confirming that the model is insensitive to parameter deviations. To validate the applicability of the logistic function, we compared it with three alternative models—linear, exponential, and power-law functions (Table 3). The results show that the logistic model achieved the highest R^2 (0.686), outperforming the linear ($R^2 = 0.616$), exponential ($R^2 = 0.586$), and power-law ($R^2 = 0.652$) models. Furthermore, the logistic model yielded the lowest AIC value ($AIC = -128,264.29$), indicating that it achieved the best balance between goodness-of-fit and model complexity. These results confirm that the

logistic function can more effectively capture the nonlinear stress effects of water-salt factors on vegetation cover.

After computing the WSS index from 1985 to 2025 using Eq. (3-1) and applying multi-year normalization to all image pixels, we obtained the spatiotemporal distribution characteristics of WSS in the Yellow River Delta (Fig. 3 a-i). Spatially, areas near the inland Yellow River channel experience relatively low water-salt stress ($WSS < 0.4$), while regions closer to the sea that are susceptible to tidal inundation show significantly higher water-salt stress ($WSS > 0.7$). The mean value of WSS within the nature reserve was 0.61, with a standard deviation of 0.138. The mean value of WSS outside the nature reserve was 0.54, with a standard deviation of 0.087. This indicates that water-salt stress is stronger within the nature reserve than outside, with greater spatial variability observed in the nature reserve. From 1985 to 2025, the Diaokou River Nature Reserve (DKNR) exhibited continuous coastal erosion, accompanied by a noticeable expansion of areas affected by high water-salt stress. In the Qingshuigou Nature Reserve (QSGNR), the river mouth spit extended southeastward from 1985 to 1995, coinciding with an expansion of regions with lower WSS values. Following the 1996 diversion of the Yellow River channel from the southern to the northern branch, the northern spit advanced progressively northeastward, and the low-WSS zone expanded correspondingly. This newly formed land, together with improved water-salt conditions, has fostered better growing environments for wetland vegetation and enhanced ecological condition. The spit of the southern branch, in the QSGNR has experienced localized shoreline erosion due to the lack of water and sediment supply, and the low-WSS zones has decreased.

Comparison between RSEI and WSRSEI models

Comparison of internal structures of RSEI and WSRSEI

Following the aforementioned method in Eq. (3-2), we selected four indexes (NDVI, WET, LST, and NDBSI) for PCA and applied multi-year normalization to compute the RSEI. Correlation analysis between RSEI and the annual values of WET, NDVI, NDBSI, LST, and SI was then conducted (Fig. 4a-d and i). The results indicate that NDVI exhibited a strong positive correlation with RSEI ($R^2 = 0.83$), whereas WET showed a negative correlation ($R^2 = 0.34$). SI was also negatively correlated with RSEI ($R^2 = 0.33$), while NDBSI and LST demonstrated weaker correlations.

Notably, while most previous studies have reported a positive relationship between WET and RSEI (Zhang et al. 2022; Xiong et al. 2021), our findings in the estuarine delta region revealed an inverse relationship. This discrepancy

Table 3 Comparison of the performance of different model structures in fitting the relationship between WSS-FVC

Model form	Formula	R^2	AIC
Linear model	$WSS = a \cdot SI + b \cdot WET + c$	0.616	-120,690.84
Exponential model	$WSS = a \cdot e^{(b \cdot SI + c \cdot WET)}$	0.586	-117,870.49
Power-law model	$WSS = a \cdot SI^b \cdot WET^c$	0.652	-124,391.83
Logistic model	$WSS = a / (1 + e^{(b \cdot SI + c \cdot WET)})$	0.686	-128,264.29

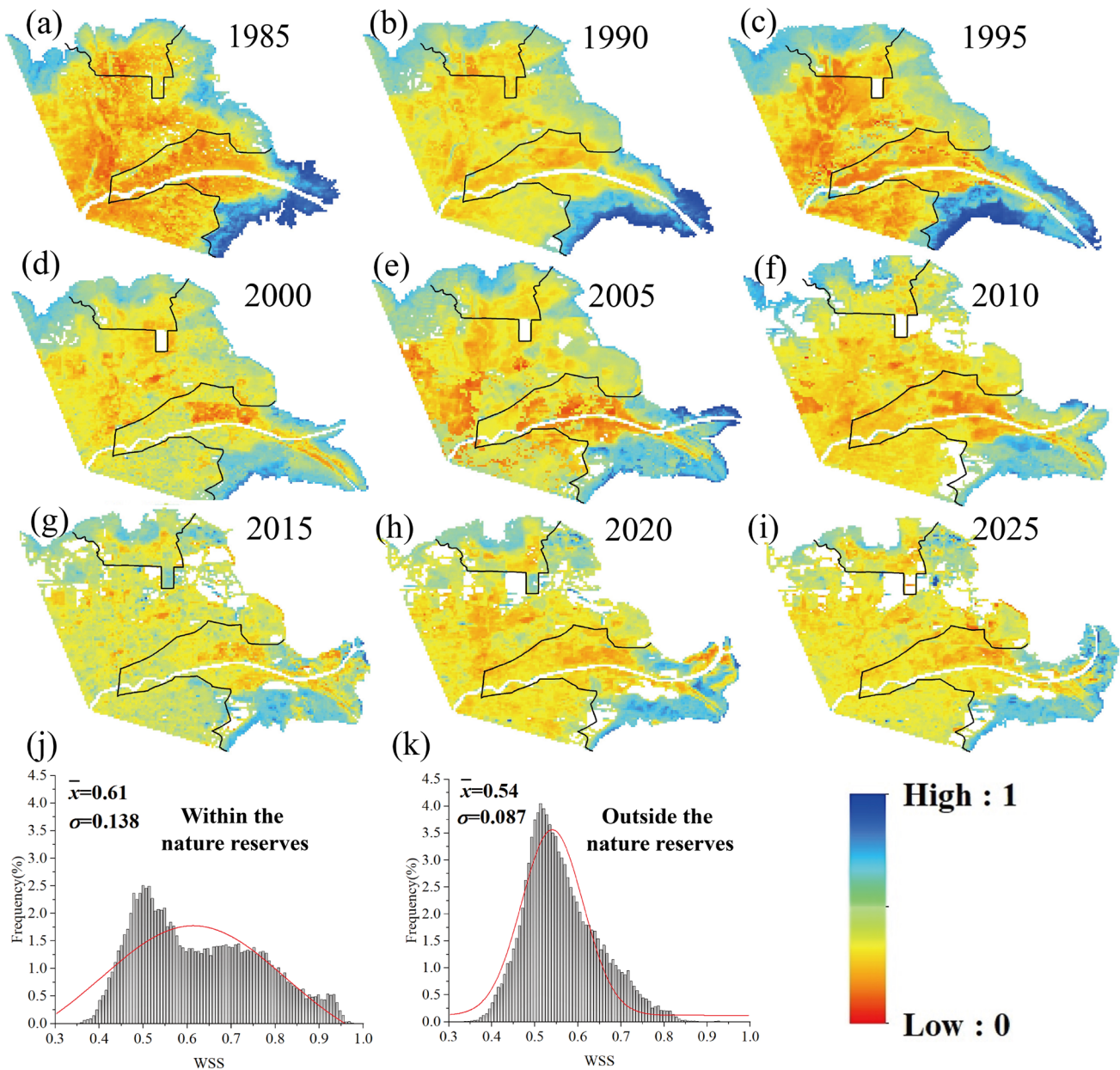


Fig. 3 Spatiotemporal distribution characteristics of the WSS in the Yellow River Delta: (a) 1985, (b) 1990, (c) 1995, (d) 2000, (e) 2005, (f) 2010, (g) 2015, (h) 2020 and (i) 2025. The histogram of WSS (j) within the nature reserves and (k) outside the nature reserves

can be attributed to the high salinity content in soil moisture, which inhibits vegetation growth. Consequently, the conventional practice of directly integrating WET into RSEI construction is unsuitable for accurately evaluating ecological condition in estuarine settings.

The WSRSEI was obtained by conducting PCA according to Eq. (3–3) on four indices (WSS, NDVI, NDBSI, and LST), followed by multi-year normalization. By establishing multi-year (1985–2025) correlation relationships between WSS, NDVI, NDBSI, LST and WSRSEI (Fig. 4e–h), we found that the WSS showed a strong negative correlation

with WSRSEI ($R^2=0.85$). NDVI exhibited a positive correlation with RSEI ($R^2=0.82$), while NDBSI and LST demonstrated weaker correlations. This indicates that WSS dominates the first principal component of WSRSEI.

Throughout the entire Yellow River Delta, WSRSEI is slightly larger than RSEI, and the Mean Absolute Difference (MAD) value is 0.033; In areas with low WSS ($WSS < 0.4$), WSRSEI values exceeded RSEI values (Fig. 5), with a mean difference of 0.01. Conversely, in regions with high WSS ($WSS > 0.7$), WSRSEI values were lower than RSEI values, showing a mean difference of -0.041 . Moreover,

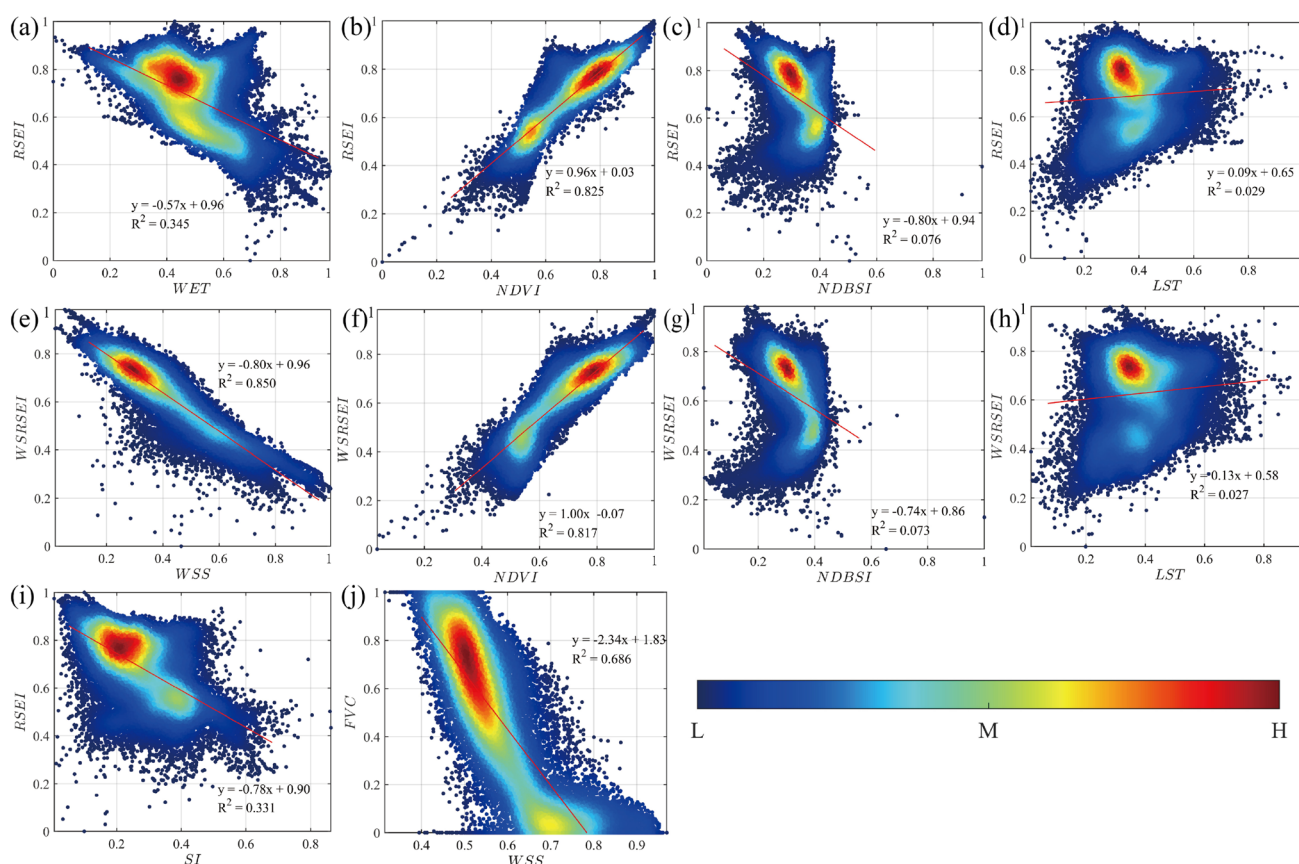


Fig. 4 Correlation relationships between the indexes during 1985–2025. **a** WET vs.RSEI; **b** NDVI vs.RSEI; **c** NDBSI vs.RSEI; **d** LST vs.RSEI; **e** WSS vs.WSRSEI; **f** NDVI vs.WSRSEI; **g** NDBSI vs.WSRSEI; **h** LST vs.WSRSEI; **i** SI vs.RSEI; **j** WSS vs.FVC

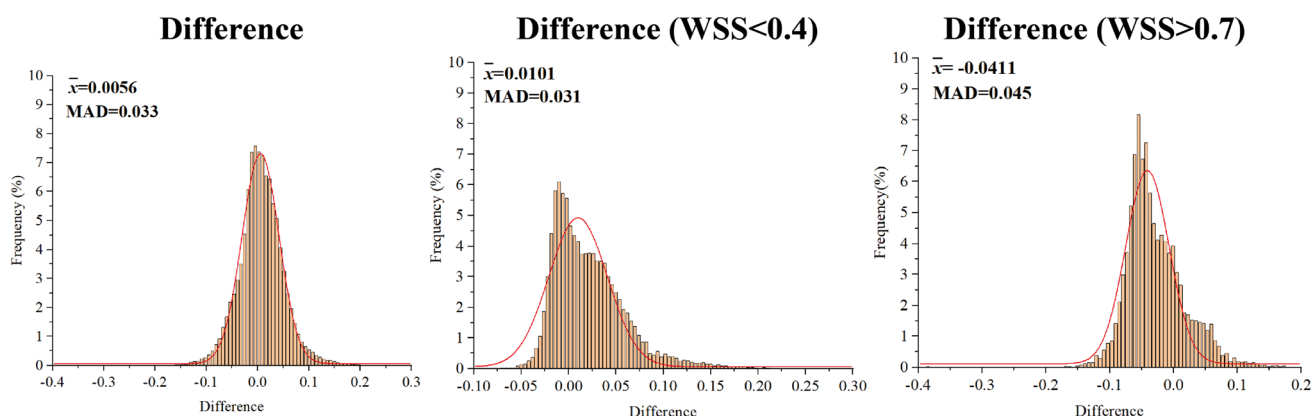


Fig. 5 The histogram of differences between WSRSEI and RSEI during 1985–2025. **a** Entire Yellow River Delta; **b** Low-WSS areas; **c** High-WSS areas

the MAD between the two indices was greater in high-WSS areas than in low-WSS areas (Fig. 5). These results indicate that the discrepancies between WSRSEI and RSEI are more pronounced under conditions of high water-salt stress. However, The ecological merits and demerits of WSRSEI and RSEI need to be verified through independent field measurement data.

Independent model validation using field-measured data

To determine which of the observed structural differences between RSEI and WSRSEI (Section "Comparison of Internal Structures of RSEI and WSRSEI") better reflects actual ecological conditions, we used field-measured data that were entirely independent of the model construction and

parameter calibration processes to evaluate the assessment accuracy of both models against real vegetation conditions.

(1) Remote sensing retrieval accuracy of SI and WET

The measured soil salinity (SSC) and soil moisture content (SMC) from the 40 sampling sites were used to validate the retrieval accuracy of SI and WET, respectively. The results show that SI is significantly positively correlated with measured EC ($R^2=0.711$, $RMSE=1.07$ dS/m, $p<0.05$) (Fig. 6a), confirming the effectiveness of SI for soil salinity retrieval in the Yellow River Delta. Linear regression between measured soil gravimetric moisture content (SMC) and the corresponding WET values (Fig. 6b) indicates a moderately positive correlation ($R^2=0.731$, $RMSE=0.018$, $p<0.05$), confirming the validity of the selected input indicators in the Yellow River Delta.

(2) Comparison of assessment accuracy between WSRSEI and RSEI

The measured above-ground biomass (AGB, g/m^2) from the 40 vegetation quadrats was regressed against the corresponding RSEI and WSRSEI values, respectively (Fig. 6c,d). The results show that RSEI is positively correlated with measured biomass ($R^2=0.557$, $RMSE=85.044$ g/m^2 , $p<0.05$), indicating that the conventional model can reflect vegetation conditions to some extent, but with limited accuracy. In contrast, WSRSEI shows significantly higher correlation with measured biomass ($R^2=0.624$, $RMSE=78.34$ g/m^2 , $p<0.05$), representing a 6.7% improvement in R^2 and a 7.9% reduction in RMSE. This confirms that the structural adjustment of WSRSEI does indeed yield a genuine improvement in ecological assessment accuracy.

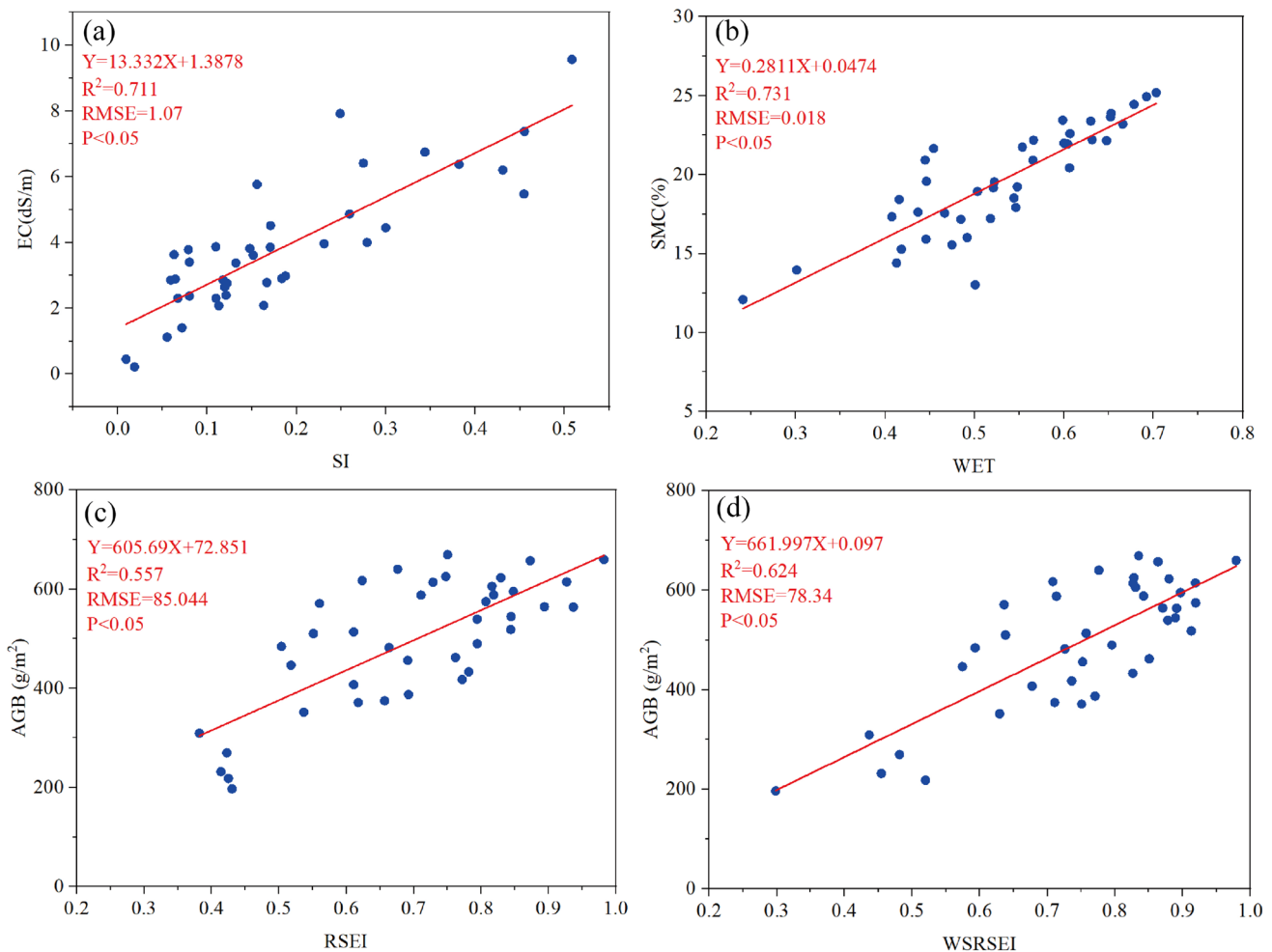


Fig. 6 Independent field validation. **a** SI vs. measured soil salinity; **b** WET vs. soil moisture content; **c** RSEI vs. measured average vegetation biomass; **d** WSRSEI vs. measured average vegetation biomass

Ecological condition assessment of the Yellow River Delta

To assess the statistical reliability of the long-term trends, we quantified the methodological uncertainties in model construction from three aspects: PCA loading interannual variability (U_{PCA}), GA parameter optimization uncertainty (U_{GA}), and multi-sensor consistency deviation (U_{sensor}).

To evaluate the interannual stability of the PCA structure, we compiled the annual PC1 loading coefficients for 1985–2025 (Supplementary Table S1). The results show that the standard deviations of the loading coefficients for NDVI, WSS, NDBSI, and LST were 0.0125, 0.0079, 0.0095, and 0.0057, respectively. The average of these four standard deviations was 0.0089, which was taken as the interannual variability of PCA loadings (U_{PCA}), confirming that the WSRSEI model structure exhibited good stability over the 40-year time series.

For the uncertainty associated with GA parameter optimization, we evaluated this by independently running the GA 20 times with different random initial populations. The results (Supplementary Table S2) show that the R^2 values from all 20 runs were identical (0.686), with no detectable variation, confirming the high convergence stability of the GA optimization. Accordingly, U_{GA} was estimated to be less than 0.001, which is negligible in the composite standard error.

For the uncertainty arising from multi-sensor consistency deviation, we compared the WSRSEI results derived from Landsat 5 TM, Landsat 7 ETM+, and Landsat 8 OLI. The coefficients of determination (R^2) for WSRSEI between any pair of sensors were all above 0.96, indicating high consistency among multi-sensor data and supporting their direct concatenation for long-term time series analysis (Supplementary Fig. S1). Based on the overlapping image pairs, the standard deviation of the annual mean WSRSEI differences between L5 and L7 was 0.002, and between L7 and L8 was 0.006. The average of these two values was taken as the estimate of multi-sensor consistency deviation, yielding $U_{sensor} \approx 0.004$.

Based on the PCA loading interannual variability (U_{PCA}), GA parameter optimization uncertainty (U_{GA}), and multi-sensor consistency deviation (U_{sensor}), the composite standard error (SE) of the annual mean WSRSEI was synthesized using the root-sum-square method, yielding $SE \approx \pm 0.0129$.

Based on the methods outlined above, the spatial distribution of the WSRSEI was calculated across the Yellow River

Delta for the period from 1985 to 2025. All pixel values from all years were normalized to the range of 0 to 1. The WSRSEI values, representing ecological condition, were then classified into five grades—"Poor," "Fair," "Moderate," "Good," and "Excellent" (Table 4)—using the Jenks Natural Breaks method.

The results show that regions adjacent to inland areas and along river channels serve as the primary zones of "Excellent" and "Good" ecological condition, with particularly favorable conditions observed along both sides of the active Qingshuigou channel. In contrast, coastal areas subject to seawater inundation are dominated by "Poor" and "Fair" ecological condition grades (Fig. 7).

Between 1985 and 2025, the WSRSEI values exhibited a slight overall decreasing trend, with mean values consistently ranging from 0.61 to 0.64 (Fig. 8a), indicating that ecological condition remained relatively stable across the Yellow River Delta and maintained an overall "Moderate" level throughout the study period. The standard deviation, varying between 0.145 and 0.158, showed no significant change in data dispersion over time. At the entire delta scale, the mean WSRSEI decreased from 0.631 in 1985 to 0.621 in 2025, representing a change of 0.010. This change is less than twice the standard error ($2 \times SE = 0.0258$), indicating that the declining trend in ecological condition across the entire Yellow River Delta during 1985–2025 is not statistically significant ($p > 0.05$), and that the overall ecological condition of the region remained relatively stable.

As areas under prioritized protection by management authorities, nature reserves are subject to relatively limited human impact. However, due to intense erosion and deposition dynamics, the ecological condition in these regions has undergone significant changes. The overall ecological condition showed a slight decline before 2008, with the WSRSEI decreasing from 0.582 in 1985 to 0.574 in 2008. In contrast, the period after 2008 saw a significant and sustained improvement, with the index reaching 0.622 by 2025 (Fig. 8b)—representing an upgrade from "moderate" to "good" ecological condition. For the entire Nature Reserve, WSRSEI increased from 0.582 to 0.622, with the change magnitude exceeding twice the standard error, confirming that the improving trend in ecological condition within the nature reserve is statistically significant and unambiguous.

The QSGNR exhibited a consistent improvement in ecological condition throughout the study period. Between 1985 and 2025, the progressive progradation of the spit along the Qingshuigou channel was accompanied by a substantial increase in ecological condition, reflected in a rise of the WSRSEI from 0.565 to 0.627 (Fig. 8c), the change magnitude exceeding twice the standard error, confirming that the improving trend in ecological condition within the QSGNR is statistically significant. Following the

Table 4 Ecological condition classification

WSRSEI	0–0.40	0.40–0.53	0.53–0.64	0.64–0.75	0.75–1
Ecological condition	Poor	Fair	Moderate	Good	Excellent

Fig. 7 Spatiotemporal distribution characteristics of WSRSEI in the Yellow River Delta: **a** 1985, **b** 1990, **c** 1995, **d** 2000, **e** 2005, **f** 2010, **g** 2015, **h** 2020 and **i** 2025

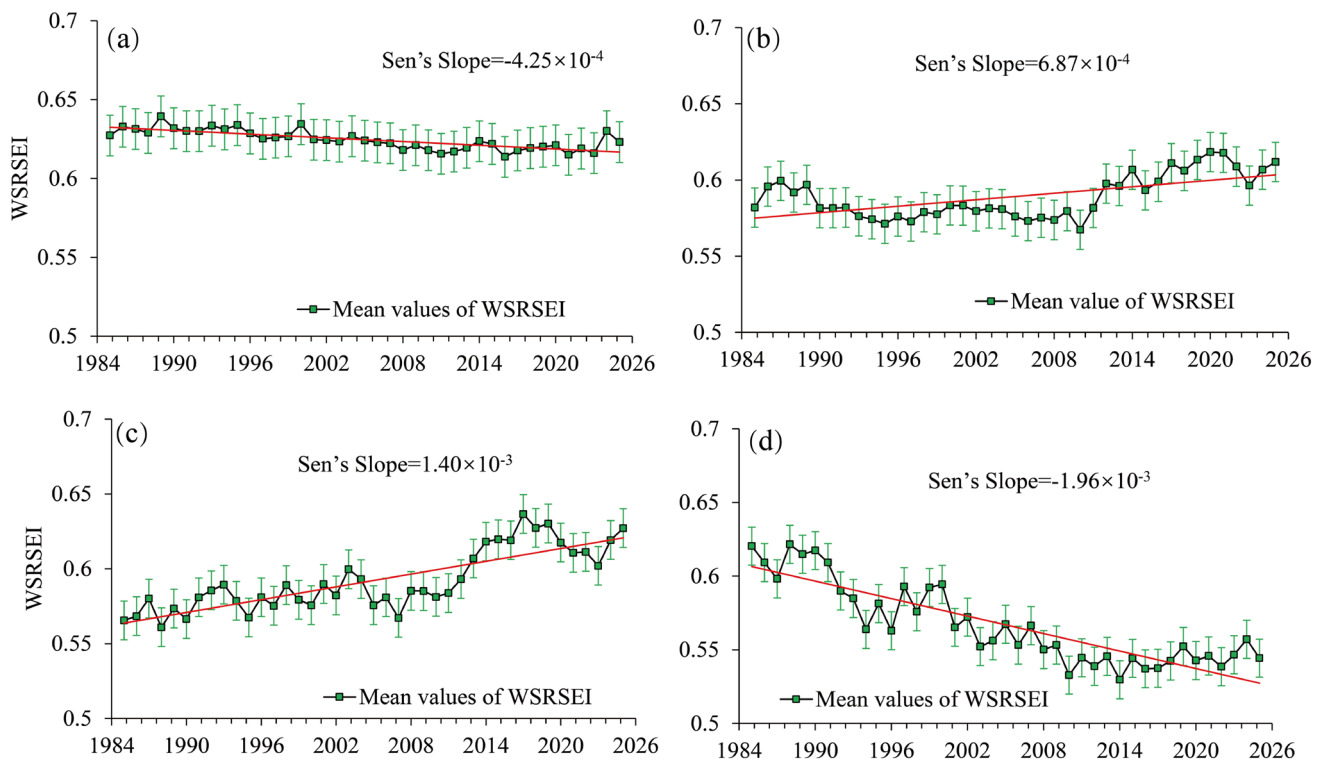
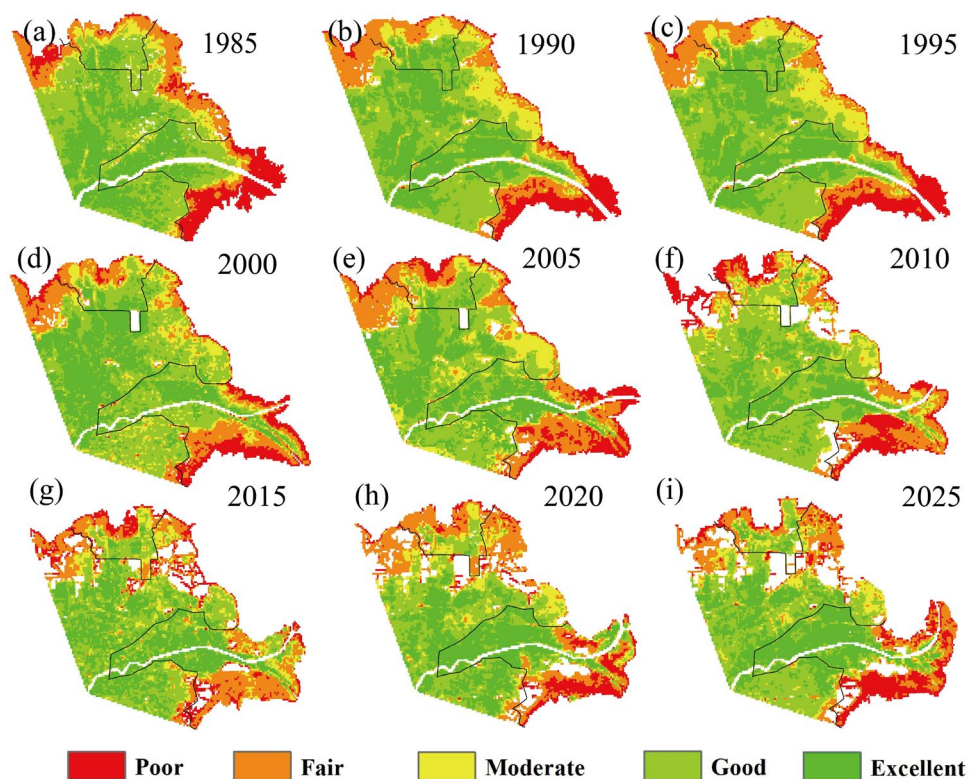


Fig. 8 Temporal variations in ecological condition: **a** Yellow River Delta, **b** Entire Nature Reserve, **c** QSGNR, **d** DKRNR

Table 5 WSRSEI trend analysis and Hurst index statistics

Region	Z-statistic	Significance	Sen's Slope	Hurst index	Future trend
Yellow River Delta	-5.51	significant downward ($p < 0.05$)	-4.25×10^{-4}	0.542	Continuous deterioration
Entire Nature Reserve	+3.25	significant upward ($p < 0.05$)	6.87×10^{-4}	0.968	Continuous improvement
QSGNR	+5.66	significant upward ($p < 0.05$)	1.40×10^{-3}	0.948	Continuous improvement
DKRNR	-6.02	significant downward ($p < 0.05$)	-1.964×10^{-3}	0.781	Continuous deterioration

abandonment of the southern branch and the activation of the northern branch in 1996, the southern spit underwent erosion, which resulted in habitat degradation. However, areas confined within the farm dikes of the southern branch retained relatively favorable habitat conditions (Fig. 7e-h). Concurrently, the northeastward advancement of the active northern branch facilitated the expansion of areas classified as "Moderate" or higher in ecological condition. After 2010, the improvement in ecological condition accelerated markedly within the QSGNR (Fig. 8c).

In contrast, the DKRNR, lacking sediment and freshwater inputs, suffered continuous coastal erosion and areal reduction (Fig. 7). Correspondingly, ecological condition exhibited persistent degradation from 0.621 in 1985 to 0.554 in 2025 (Fig. 8d), though the rate of deterioration has shown signs of gradual attenuation in recent years. The change magnitude exceeding twice the standard error, confirming that the deteriorating trend in ecological condition within the DKRNR is statistically significant.

The M-K trend test was applied to the WSRSEI time series from 1985 to 2025 (Table 5). At the entire delta scale, the Z-statistic was -5.51 ($|Z| > 1.96$) and Sen's slope was -4.25×10^{-4} , indicating a significant downward trend ($p < 0.05$), suggesting a persistent ecological decline during 1985–2025. However, the change from the initial value (0.631) to the final value (0.621) was only 0.010, which is less than twice the composite standard error ($2 \times SE = 0.0258$), indicating that this change magnitude has not exceeded the systematic error range of the methodological framework. Therefore, although a weak declining trend

is statistically detectable, the ecological condition of the entire Yellow River Delta has not experienced a substantial or significant degradation in terms of magnitude. For the entire Nature Reserve, the Z-statistic was +3.25 (> 1.96) and Sen's slope was 6.87×10^{-4} , indicating a significant upward trend ($p < 0.05$). For the QSGNR, $Z = +5.66$ (> 1.96) and Sen's slope = 1.40×10^{-3} , indicating a significant upward trend ($p < 0.05$). For the DKRNR, $Z = -6.02$ (< -1.96) and Sen's slope = -1.964×10^{-3} , indicating a significant downward trend ($p < 0.05$). The Hurst exponent analysis showed that the Hurst exponents for the entire Yellow River Delta, the entire Nature Reserve, the QSGNR, and the DKRNR were all greater than 0.5, indicating that each region will continue its existing trend.

To reveal the evolutionary characteristics of the WSRSEI spatial pattern, we selected nine key years—1985, 1990, 1995, 2000, 2005, 2010, 2015, 2020, and 2025—for spatial autocorrelation analysis. The global spatial autocorrelation results show that Moran's I was positive for all years, indicating that positive spatial autocorrelation was the dominant pattern, with similar values (high-high or low-low) consistently clustered in space. The Z-scores for all years were well above the critical value of 1.96, and all p-values were < 0.01 , indicating that the spatial autocorrelation was highly statistically significant across all years and that the clustering patterns were not random but real. From 1985 to 1995, Moran's I remained high (0.93–0.94), indicating strong clustering intensity. In 2000, Moran's I showed its first notable decline (0.876), indicating a weakening of clustering intensity. From 2005 to 2025, Moran's I fluctuated between 0.83 and 0.91, indicating that although clustering intensity varied, it remained relatively strong overall, though it did not return to the peak levels observed before 2000. The Z-score decreased from 174.78 in 1985 to 155.11 in 2025, indicating a slight decline in the significance strength of spatial autocorrelation (Table 6).

The local spatial autocorrelation (LISA) analysis results show that high-high clusters were consistently distributed along both sides of the river channels and in inland areas, with their area expanding from 301 km² in 1985 to 754 km² in 2025. Low-low clusters were mainly distributed in areas near the sea, with their area expanding from 356.25 km² in 1985 to 591 km² in 2025 (Fig. 9). These results confirm, from a spatial perspective, the roles of sediment supply, ecological water supplementation, channel diversion, and water-salt stress in shaping the spatial pattern of ecological condition.

Table 6 Global Moran's I index variation

	1985	1990	1995	2000	2005	2010	2015	2020	2025
Global moran'I	0.944	0.945	0.938	0.876	0.891	0.906	0.835	0.884	0.889
Z value	174.780	182.473	184.534	168.497	171.239	160.841	151.199	156.699	155.105
P value	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01

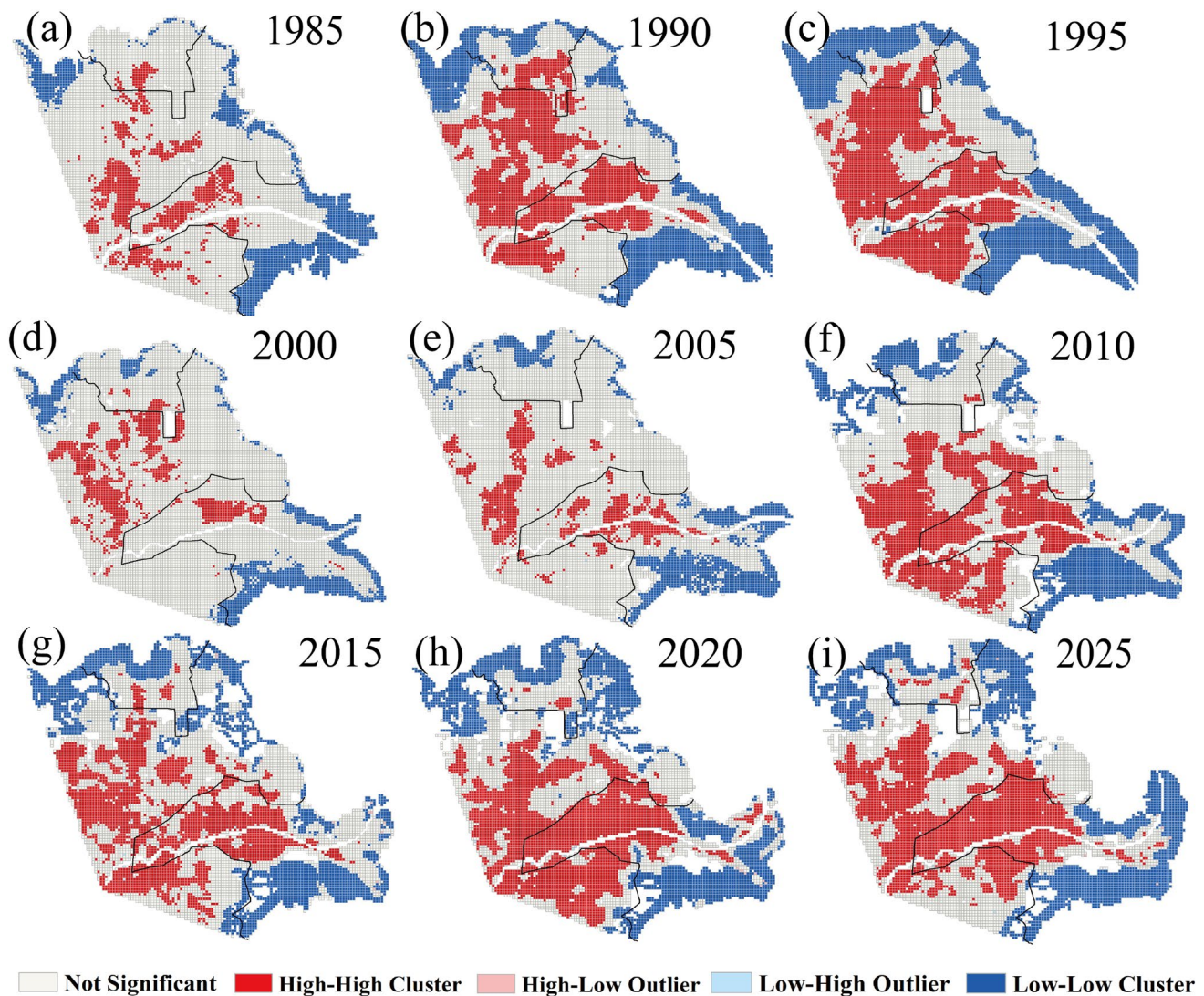


Fig. 9 Spatial autocorrelation analysis of habitat quality in the Yellow River Delta from 1985 to 2025

Discussion

Impacts of Spit dynamics & Water Supplementation on wetland habitats in nature reserves

The growth of wetland vegetation in the Yellow River Delta depends on two critical factors: growing space (Huang et al. 2022a, b) and growing environment (Yu et al. 2014). The two nature reserves within the delta—QSGNR and DKRNR—are situated in zones undergoing the most dynamic erosion and deposition processes, where changes in land area directly influence the available growing space for wetland vegetation. Regarding the growing environment, soil moisture and salinity play crucial roles in determining wetland ecological condition. To better provide the freshwater necessary for wetland vegetation growth, the Yellow River Conservancy Commission has implemented

ecological water supplementation projects specifically for the Yellow River Delta Nature Reserve: (1) Ecological water supplementation for the QSGNR. Since 2008, tailored water supplementation has been carried out for wetlands within QSGNR (Yu et al. 2021). From 2008 to 2025, ecological water supplementation was performed annually except in 2016, with a total volume reaching 1150 million m³ and an average annual supplementation of approximately 67.65 million m³ (Fig. 10d). (2) Ecological water supplementation for the DKRNR (Chi et al. 2018). Between 2010 and 2025, excluding 2016 when no supplementation occurred, the cumulative water supplementation for the Diaokou River and its terminal wetlands amounted to 440 million m³, with an average annual supplementation of nearly 29.33 million m³. (Fig. 10d).

During 1985–2025, the QSGNR received continuous sediment supply, with an average annual sediment discharge

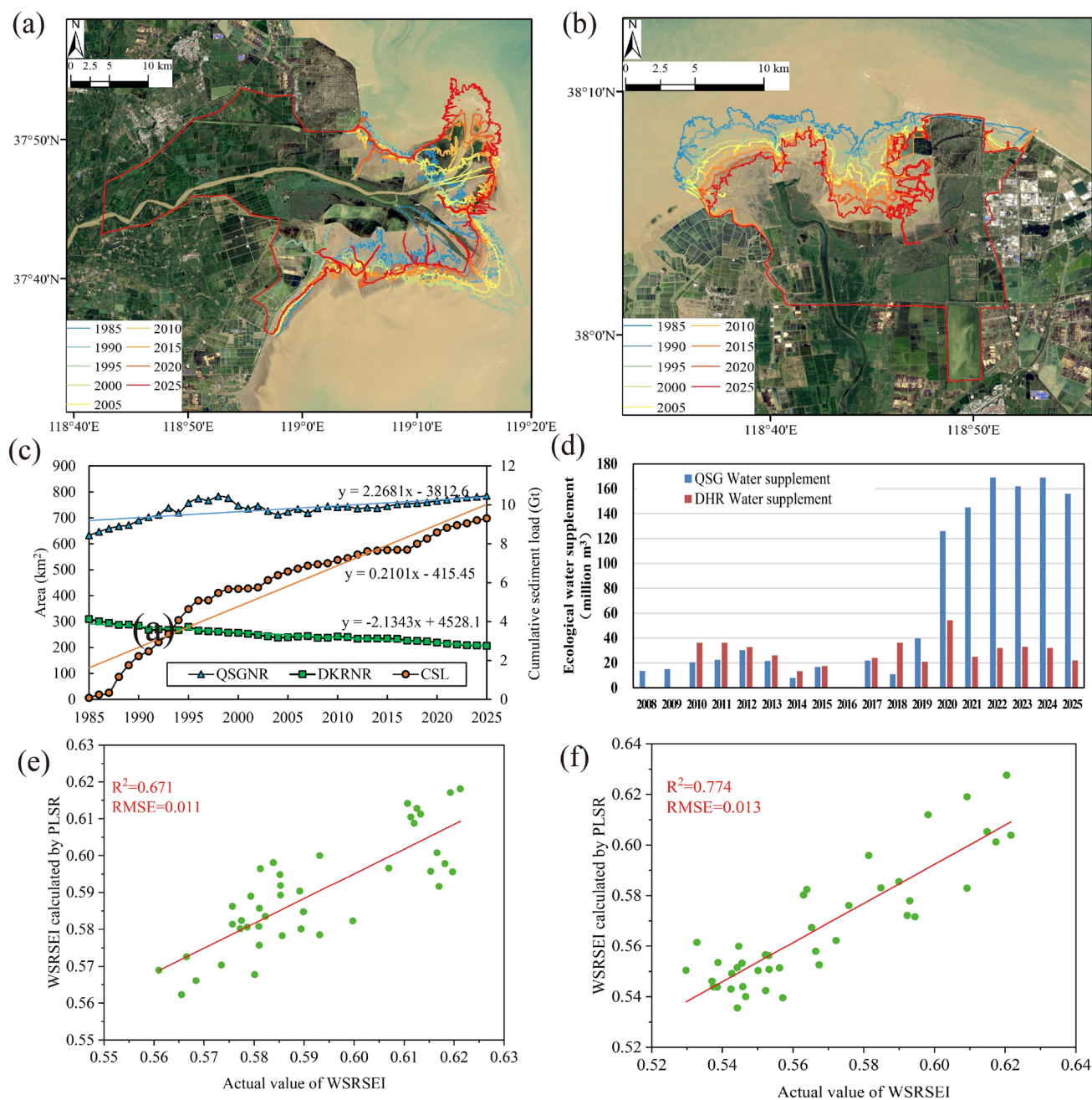


Fig. 10 **a** The change in the shoreline of the spit in the QSGNR; **b** The change in the shoreline of the spit in the DKRNR; **c** Changes in the land area of the QSGNR and DKRNR and the cumulative sediment load; **d** Ecological water supplementation volumes for the QSGNR

of 0.21 Gt/yr at the Qingshuigou estuary (Fig. 10c), accumulating a total sediment input of 9.31 Gt. This resulted in an average annual expansion of the reserve's area by 2.27 km²/yr (Fig. 10c). Following the 1996 Qing-8 diversion project, the southern branch channel, cut off from water and sediment inputs, underwent continuous shoreline erosion and

and DKRNR in the Yellow River Estuary. **e** Scatter plot of measured WSRSEI values vs. PLSR predicted values of QSGNR, **f** Scatter plot of measured WSRSEI values vs. PLSR predicted values of DKRNR

associated degradation in ecological condition. In contrast, the northern branch channel, serving as the current active distributary, demonstrated progressive shoreline progradation and sustained ecological condition improvement due to sustained water and sediment supply (Fig. 10a). Comparative analysis of areal changes and WSRSEI temporal

variations within the QSGNR revealed that improvements in ecological condition lagged behind geomorphological expansion (Fig. 8c; Fig. 10c). This delay suggests an adaptive period is necessary for wetland vegetation to colonize and establish on newly formed land. Notably, since the introduction of ecological water supplementation in 2008, the rate of ecological condition improvement has accelerated markedly, highlighting the synergistic benefits of sustained water and sediment supply and targeted water management practices.

Around the DKRNR, the absence of water and sediment supply has caused persistent coastal erosion (Fig. 10b), leading to a corresponding reduction in the nature reserve's area at an average rate of 2.13 km²/yr (Fig. 10c). Consequently, ecological condition declined continuously between 1985 and 2010. However, since 2010, the rate of ecological condition degradation has moderated, due to a gradually slowing erosion rate of the Diaokou River spit and the implementation of ecological water supplementation to the DKRNR via the Diaokou River channel (Fig. 8d). Nevertheless, as the area of the DKRNR continues to diminish, the habitat available for wetland vegetation keeps shrinking, and the overall declining trend in ecological condition remains unaltered. Therefore, to sustain ecological condition in the Diaokou River area, it is essential to first supply sufficient sediment to mitigate coastal erosion and stabilize the current extent of the nature reserve, thereby securing growing space for wetland vegetation. Subsequently, increasing ecological water supplementation will be critical to alleviating the stress imposed by seawater on the wetland ecosystem.

To quantify the synergistic effects of sediment and freshwater supply on ecological condition in the Qingshuigou (QSGNR) and Diaokou River (DKRNR) nature reserves, we established a Partial Least Squares Regression (PLSR) model using annual land area and annual ecological water supplementation volume as independent variables, and annual mean WSRSEI as the dependent variable for the period 1985–2025. The PLSR models for the two reserves achieved R^2 values of 0.671 and 0.774, and Q^2 values of 0.609 and 0.679, respectively (Table 7). The scatter plots of measured versus PLSR-predicted values (Fig. 10 e,f) demonstrate that the models have good explanatory power and predictive capability. However, the temporal coincidence between the onset of ecological water supplementation in 2008 and the subsequent improvement in WSRSEI in the

QSGNR. This effect is difficult to isolate from the concurrent influence of sediment supply.

In contrast, ecological condition change in the DKRNR was predominantly driven by land area alone (contribution: 80.83%, VIP=1.14), with water supplementation playing a limited role (contribution: 19.17%, VIP=0.84). However, because the DKRNR has experienced continuous coastal erosion since the channel diversion in 1976, its land area has been steadily shrinking. Despite both standardized coefficients being positive (1.03 and 0.25), WSRSEI has continued to decline. This indicates that, in the absence of sustained sediment supply, land area loss is the fundamental driver of habitat degradation, and limited water supplementation cannot offset the negative effects of spatial loss.

Effects of engineered geomorphic structures on wetland ecological condition

Following the 1976 channel diversion, during which the Qingshuigou course superseded the Diaokou River course as the Yellow River's primary discharge pathway, relevant authorities progressively constructed and reinforced levees along the channel to ensure flood control safety. The left bank levees span 49.731 km, while the southern right bank levees extend 27.735 km. Current estuary levees typically feature crest elevations of 5–7 m, crest widths of 6–8 m, and side slopes ranging from 1:2.5 to 1:3.0. By the 1980s, in response to severe channel shrinkage and rising water levels within the discharge pathway, farm dikes were constructed to consolidate flow distribution, concentrate water discharge, enhance sediment transport capacity, and promote sediment flushing to the sea. However, these structures have also accelerated the development of a perched river system (Du et al. 2024). The Gudong Sea embankment, completed in 1988 after three years of construction, was engineered to ensure the safe extraction of the Gudong Oilfield. Extending approximately 17 km along the coast, it serves as a formidable barrier against storm surges, seawater erosion, and flooding from the Yellow River. By shielding the Gudong Oilfield, surrounding farmland, and adjacent residential communities, the sea embankment has come to be regarded as the "lifeline" of the oilfield's operations.

Analysis of cross section datas, integrated with WSS and WSRSEI values from 2015 and 2025 (Fig. 11), shows that in Cross Section 1, areas inside the levees exhibit an average topographic elevation 1.21–1.45 m higher, WSS

Table 7 The PLSR simulation results

Region	Driving factor	Standardized regression coefficient	VIP value	Contribution (%)	R^2	Q^2
QSGNR	land area	0.48	1.01	53.74	0.671	0.609
	Water supplementation volumes	0.41	0.99	46.26		
DKRNR	land area	1.03	1.14	80.83	0.774	0.679
	Water supplementation volumes	0.25	0.84	19.17		

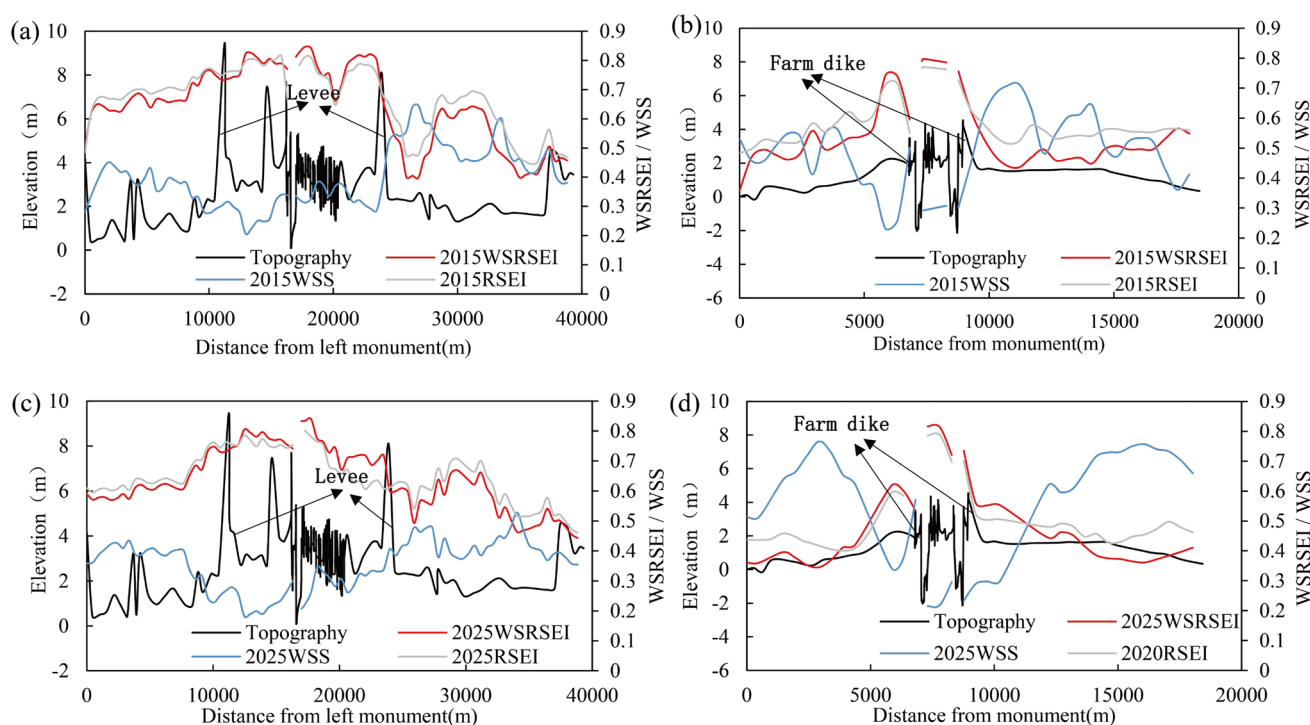


Fig. 11 Cross-sectional distribution characteristics of WSS, WSRSEI and topography: **a** Cross Section 1 in 2015; **b** Cross Section 2 in 2015; **c** Cross Section 1 in 2025; **d** Cross Section 2 in 2025

Table 8 Spatial variations of topography, WSS, RSEI and WSRSEI along cross sections in 2015 and 2025

Cross Section	Zone	Elevation	2015			2025		
			WSS	RSEI	WSRSEI	WSS	RSEI	WSRSEI
Cross Section 1	Left floodplain	1.37	0.339	0.720	0.705	0.321	0.685	0.674
	Floodplain within the levees	2.82	0.293	0.754	0.781	0.303	0.676	0.735
	Right floodplain	1.61	0.498	0.567	0.511	0.418	0.591	0.554
Cross Section 2	Left floodplain	1.01	0.452	0.573	0.535	0.592	0.542	0.429
	Floodplain within the farm dikes	2.66	0.300	0.767	0.791	0.243	0.757	0.784
	Right floodplain	1.31	0.536	0.564	0.511	0.542	0.486	0.447

values 0.018–0.205 lower, and WSRSEI values 0.061–0.270 higher than those outside the levees. Similarly, in Cross Section 2, areas within farm dikes display an average elevation 1.35–1.65 m higher, WSS values 0.152–0.349 lower, and WSRSEI values 0.256–0.355 higher compared to adjacent exterior areas (Table 8). As can be seen from the aerial image in Fig. 1d, vegetation growth inside the production dikes is significantly better than that outside.

A direct comparison between RSEI and WSRSEI within the same transects reveals a systematic difference (Table 8). In low-WSS zones inside the levees/farm dikes, WSRSEI is slightly higher than RSEI, indicating that the salinity correction in WSRSEI does not artificially depress ecological quality where water-salt stress is low. However, in high-WSS zones outside the levees (e.g., the right floodplain of Section 1, WSS=0.418–0.498), WSRSEI (0.511–0.554) is markedly lower than RSEI (0.567–0.591). This divergence is even more pronounced in Section 2, where the

right floodplain exhibits the highest WSS (0.536–0.542), WSRSEI (0.447–0.511) is considerably lower than RSEI (0.486–0.564).

To quantitatively assess the protective effects of levees/farm dikes on local ecological condition, this study conducted independent-sample t-tests on the WSRSEI pixel values inside and outside the embankments for the years 2015 and 2025 (Table 9). The results indicate that for Section 1, the WSRSEI values inside the levees were significantly higher than those outside ($t=10.74\text{--}13.10$, $p<0.001$), with mean differences ranging from 0.169 to 0.203 and Cohen's d values from 2.476 to 2.548, far exceeding the large-effect threshold of 0.8, demonstrating that the protective effect of the embankment on ecological condition is highly significant. Moreover, the standard deviation outside the embankment (0.074–0.108) was considerably larger than that inside (0.042–0.045), indicating that areas without embankment protection exhibit greater spatial heterogeneity in WSRSEI

Table 9 Results of the independent samples T-test of WSRSEI

Cross Section		Cross Section 1		Cross Section 2	
		Inside	Outside	Inside	Outside
2015	Mean value	0.784	0.581	0.729	0.497
	Std	0.042	0.108	0.077	0.037
	T value	13.100		12.165	
	P value	<0.001		<0.001	
	Cohen's d	2.476		4.858	
2025	Mean value	0.756	0.587	0.675	0.404
	Std	0.045	0.074	0.112	0.056
	T value	10.740		9.122	
	P value	<0.001		<0.001	
	Cohen's d	2.548		3.724	

and more pronounced fluctuations in ecological condition. For Section 2, the WSRSEI values inside the farm dikes were also significantly higher than those outside ($t=9.122-12.165$, $p<0.001$), with mean differences ranging from 0.232 to 0.271 and Cohen's d values from 3.724 to 4.858, again far exceeding the large-effect threshold of 0.8, confirming that the protective effect of the farm dikes on ecological condition is equally highly significant. In addition, the standard deviation inside the farm dikes (0.077–0.112) was greater than that outside (0.037–0.056), suggesting that areas with farm dikes protection exhibit greater spatial heterogeneity in WSRSEI and more pronounced fluctuations in ecological condition. This statistical evidence corroborates the statistical significance of these structures may help mitigate water-salt stress at the local scale.

Effects of land use change on ecological condition

Driven by the natural evolution of the Yellow River estuary and continued socioeconomic development in the region, land use patterns in the Yellow River Delta have changed considerably, significantly affecting local ecological condition. Using land use data from 1985, 1995, 2010, and 2020 in the modern Yellow River Delta, this study compared ecological condition across different land use types. The results show that cropland had the highest ecological condition, with a median WSRSEI of 0.739 and an interquartile range (25th–75th percentile) of 0.695–0.779, suggesting low dispersion (Fig. 12e). Grassland and forest land showed comparable ecological condition to cropland, with a median WSRSEI of 0.724 and an interquartile range (25th–75th percentile) of 0.673–0.769, also indicating low variability. Built-up areas exhibited relatively lower ecological condition, recording a median WSRSEI of 0.492 and an interquartile range (25th–75th percentile) of 0.416–0.563, reflecting moderate dispersion. In contrast, bare land and marsh areas had the poorest ecological condition, with a median WSRSEI of only 0.325 and an interquartile range (25th–75th percentile) of 0.261–0.422, alongside high variability. In summary, cropland, grassland, and forest land are linked to higher and more stable ecological condition, while built-up areas, marsh and bare land are characterized by lower quality and greater variability.

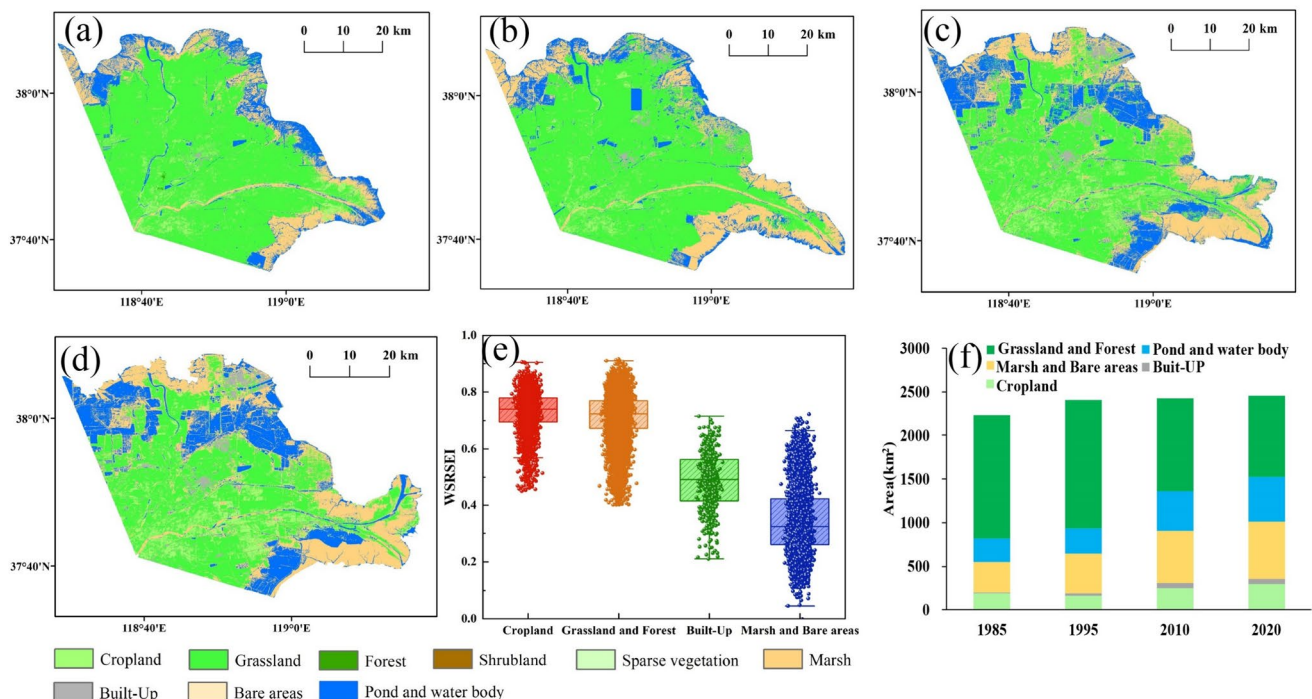


Fig. 12 Spatiotemporal evolution characteristics of land use types from 1985 to 2020: **a** 1985, **b** 1995, **c** 2010, **d** 2020, **e** WSRSEI values corresponding to different land use types, and **f** Area of land use types in the Yellow River Delta across different years

Human activities played a major role in land use changes between 1985 and 2020. Pond and water body areas increased from 271.19 km² to 509.87 km² (a 55.64% increase), built-up land expanded from 11.80 km² to 65.30 km² (a 453.16% increase), marsh and bare land grew from 350.75 km² to 655.62 km² (an 86.92% increase), and cropland rose from 188.63 km² to 293.59 km² (a 55.64% increase). In contrast, grass land and forest land experienced the most pronounced decline, decreasing from 1406.02 km² to 934.32 km² (a 33.55% reduction) (Fig. 12f).

The spatial distribution of land use types is closely linked to the spatial pattern of WSRSEI values. Cropland, grassland, and forest land, which have higher WSRSEI medians, are mainly located in the southwestern region and along river corridors. These areas benefit from freshwater influence that alleviates soil salinization. In contrast, bare land and marsh, which have lower WSRSEI values, are concentrated in the coastal zone where water-salt stress is most severe. Temporally, the expansion of ponds, bare land, and built-up areas has progressively encroached upon grassland and forest (Fig. 12a–d). These land use types are consistently associated with lower WSRSEI values, suggesting that their expansion has contributed to habitat fragmentation and degradation in regions where natural vegetation loss has been most pronounced.

It should be acknowledged that the land use data are limited to four discrete time points (1985, 1995, 2010, and 2020), which precludes formal statistical correlation analysis between land use area changes and WSRSEI trends due to insufficient sample size. The interpretation presented here is based on the convergence of spatial patterns and temporal consistency. This evidence supports the inference that land use conversion, particularly the loss of natural vegetation to artificial surfaces and bare land, has been a contributing factor to the overall degradation of ecological condition. However, this remains a qualitative inference rather than a statistically demonstrated causal mechanism.

Strengths and limitations

Strengths

In constructing the WSS index and the improved WSRSEI model, all Landsat series images were uniformly preprocessed with atmospheric correction and cloud masking. The fundamental parameters (NDVI, WET, NDBSI, LST, FVC) were calculated following standard procedures in remote sensing ecology, ensuring consistent data quality throughout the time series. The newly incorporated SI adopts a targeted inversion model that has been previously validated with field measurements in the Yellow River Delta (Fouad 2003; Cheng et al. 2022). By ecologically coupling SI with WET

through a logistic function, the proposed approach constructs a more effective composite stress indicator (WSS), thereby substantially improving the ecological representation capacity of the RSEI framework in estuarine wetland settings.

Statistical results demonstrate that WSS explains 85% of the variance in ecological condition ($R^2=0.85$), far exceeding the explanatory power of either WET ($R^2=0.345$) or SI ($R^2=0.33$) alone. Furthermore, the correlation between WSS and WSRSEI is stronger than that between WSS and its optimization target FVC ($R^2=0.686$), confirming both the effectiveness and independence of WSS in contributing to ecological condition assessment. Spatially, the distribution of WSS is independent of model calculations: high-value areas closely align with the tidally influenced coastal zone (Fan et al. 2022), while low-value areas correspond to riverine freshwater-influenced zones. This clear and physically plausible pattern provides intuitive yet compelling validation of the index's credibility. From a mechanistic perspective, the optimization results ($|b|>|c|$) explicitly identify SI as the dominant stress factor, endowing WSS with a clear ecological interpretation.

Limitations

While this study improves the scientific rigor of ecological condition assessment in estuarine regions through the introduction of the WSS, several limitations should be acknowledged.

First, regarding the transferability of the proposed framework, although the methodological concept of coupling moisture and salinity into a composite stress index (WSS) is broadly applicable to other coastal wetlands and deltaic environments, the specific parameterization of the logistic function (a , b , c) is system-dependent. In estuarine systems with different salinity regimes, different vegetation communities, or contrasting hydrological conditions, the stress-response relationship may differ substantially. Consequently, direct application of the WSS formula without recalibration to other systems would be inappropriate. However, the methodological framework is generalizable and can be adapted to other coastal wetland environments following localized calibration.

Second, both the RSEI and WSRSEI models incorporate a limited set of indicators. Important environmental and anthropogenic factors such as precipitation, evaporation, soil texture, nutrient content, and human disturbances have not yet been integrated into the current assessment framework. Future studies could enhance the explanatory power of the model by incorporating these additional variables, particularly through the integration of meteorological and soil physicochemical data.

Third, the independent validation in this study was based on a single-year field campaign (2020). Multi-year validation would further strengthen confidence in the model's temporal stability, particularly under inter-annual climate variability. Future work should prioritize establishing long-term field monitoring networks to support more rigorous validation across multiple years and seasons.

Despite these limitations, the current findings already offer a robust foundation for the methodological validity and innovation of the WSS index and the WSRSEI model, and demonstrate their potential for application in other estuarine and coastal settings following appropriate system-specific calibration.

Comparison with previous studies

To validate the reliability of our findings, we systematically compared our WSRSEI-based assessments with previous studies in the Yellow River Delta.

Regarding overall trends, Zhang et al. (2022) reported that ecological quality in the Yellow River Delta declined during 2000–2020 using the conventional RSEI. Wang et al. (2025), based on the InVEST model, revealed a fluctuating trend in habitat quality during 1980–2020. Our study shows that WSRSEI decreased from 0.631 to 0.621 over 1985–2025, which is highly consistent with the overall declining trend reported in these studies.

In terms of spatial patterns, Wang et al. (2026) found that poor ecological quality was concentrated in coastal tidal flats, while better quality was distributed in the southwestern region and along the Yellow River channel. Zhang et al. (2022) reported that high-high clusters occurred mainly in vegetated areas, whereas low-low clusters were found in built-up and coastal zones. Our WSRSEI spatial distribution is fully consistent with these findings—low values in coastal intertidal zones and high values along inland river channels—with spatial autocorrelation also dominated by high-high and low-low clustering.

Despite the consistent trends, this study advances beyond previous research in three aspects: (1) the longest temporal coverage (1985–2025, 41 years) with continuous annual monitoring rather than discrete time points; (2) the first construction of WSS by coupling moisture and salinity through a logistic function, rather than simple indicator substitution; and (3) the first implementation of independent field validation (soil salinity, moisture, and vegetation biomass) in RSEI-based studies in the Yellow River Delta, along with systematic quantification of the composite standard error (SE) from three uncertainty dimensions, significantly enhancing the reliability and reproducibility of the assessments.

Conclusion

To more scientifically assess wetland ecological condition under water-salt stress in estuarine environments, this study developed a WSS based on the Logistic function, which integrates WET and SI to quantify the impact of high-salinity moisture in the Yellow River Delta. By substituting WET with WSS, an enhanced remote sensing ecological index, WSRSEI, was established for evaluating ecological condition in the modern Yellow River Delta.

Compared to the original RSEI, WSRSEI incorporates the inhibitory effects of water-salt stress on FVC. Based on independent field validation in the Yellow River Delta, WSRSEI demonstrated better agreement with measured vegetation biomass than RSEI, indicating its improved applicability for ecological assessment in this specific estuarine delta environment. Further validation in other coastal wetland systems would be valuable to assess its broader transferability.

Terrestrial space and growing environment are critical factors sustaining wetland ecological condition. Sediment discharge from the Yellow River has driven continuous progradation of the spit in the QSGNR, thereby expanding vegetated areas and displacing water-salt stress zones seaward. In contrast, the DKRNR, deprived of sediment supply, has undergone persistent erosion, resulting in wetland loss and landward encroachment of water-salt stress. Ecological water supplementation (freshwater release) has partially improved vegetation growing conditions and elevated ecological condition. Following its implementation, the ecological condition index in the Nature Reserve increased from 0.582 to 0.622, shifting its classification from "moderate" to "good".

Wetland ecological condition is also significantly influenced by anthropogenic geomorphic modifications and land use changes. In the Yellow River Delta, engineered structures not only alter channel and coastal morphology but also enhance ecological condition by mitigating water-salt stress in vulnerable regions. Conversely, the expansion of water bodies, bare land, cropland, and built-up areas has fragmented land use patterns across the delta and encroached upon grassland and forests, leading to an overall degradation in ecological condition.

Finally, it should be noted that the WSS parameters optimized in this study are specific to the Yellow River Delta and may not be directly transferable to other estuarine systems without recalibration. The logistic function formulation and the overall methodological framework, however, are generalizable and can be adapted to other coastal wetlands by recalibrating the parameters to local vegetation-stress relationships.

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Data availability His study used Landsat satellite imagery accessed via the Google Earth Engine platform. The original Landsat data are provided by the U.S. Geological Survey (USGS) and are available at <https://earthexplorer.usgs.gov/>. Google Earth Engine is a cloud-based platform for planetary-scale geospatial analysis (Gorelick et al., 2017), available at <https://earthengine.google.com/>.

Declarations

Competing interest The authors declare no competing interests.

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